

Machine Learning

CMPSCI 589

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Introduction to Supervised Learning

The *k*-nearest neighbors algorithm

UMassAmherst

College of Information
& Computer Sciences



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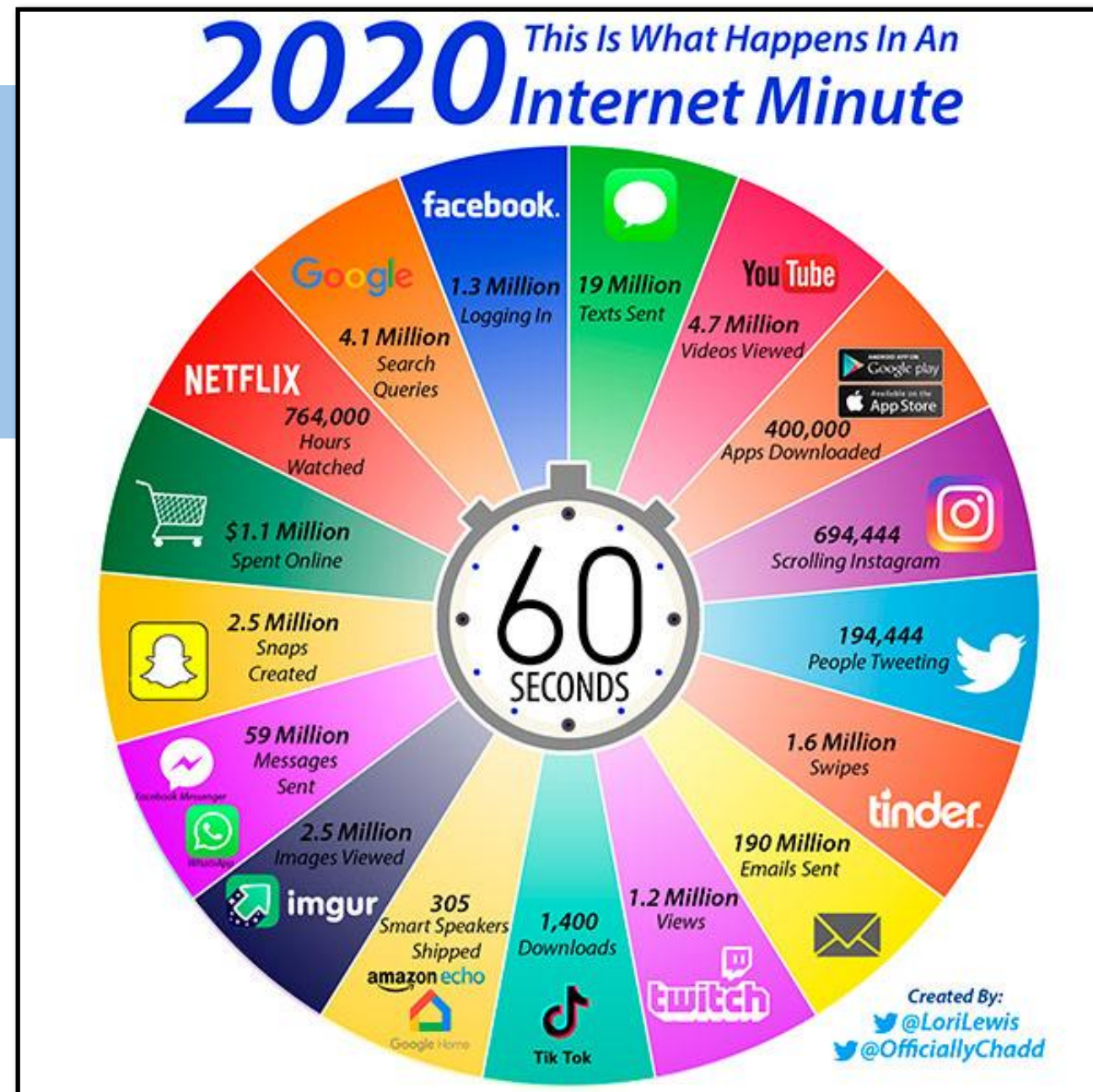


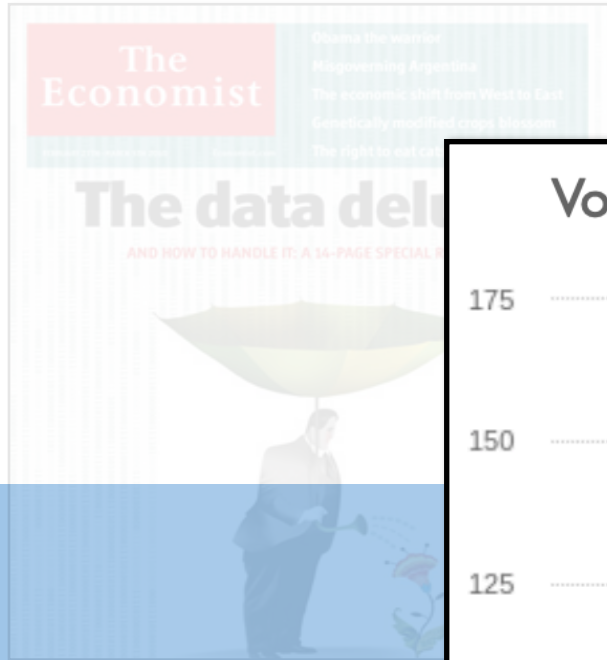
" (...) *the volume of digital data stored worldwide is set to grow 40- to 50-fold between 2010 and 2020. By the year of 2020, the digital universe will reach **44 zettabytes.*** "

Learning from data



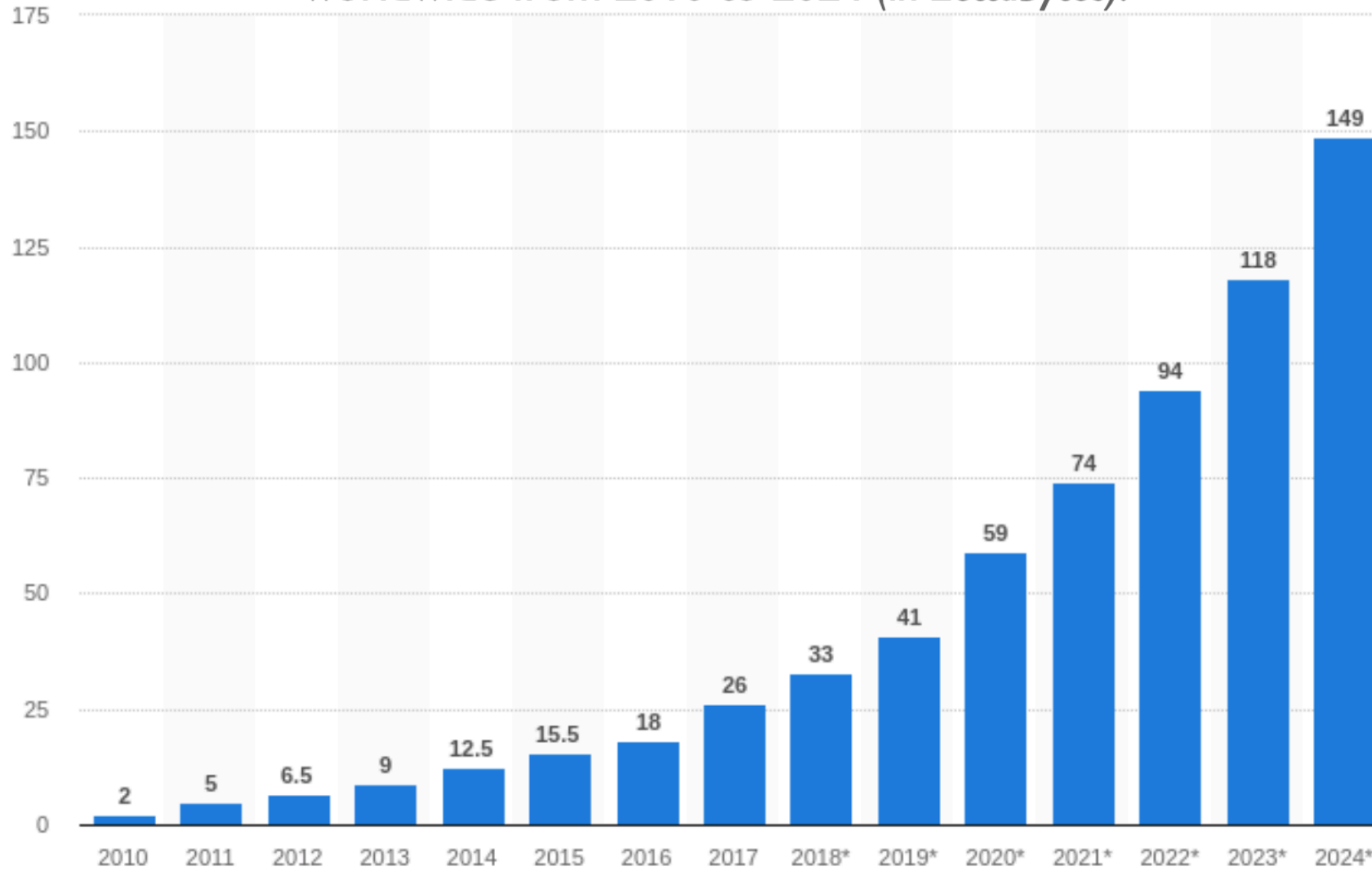
" (...)
the volume of digital data stored worldwide is set to
grow 40- to 50-fold between 2010 and 2020.
By the year of 2020, the digital universe will reach **44 zettabytes.** "





" (...) the volume of digital data stored worldwide is set to **44 zettabytes.** " (2010 and 2020.)

Volume of data/information created, captured, copied, and consumed worldwide from 2010 to 2024 (in zettabytes).



Source: Statista, 2021

2020 This Is What The Internet





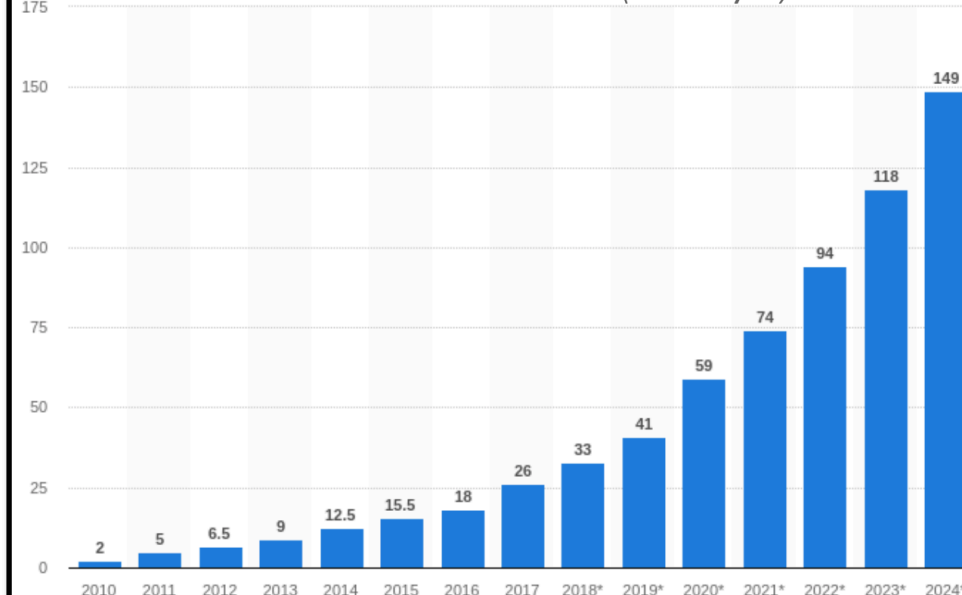
" (...)
the volume of digital data stored worldwide is set to
grow 40- to 50-fold between 2010 and 2020.
By the year of 2020, the digital universe will reach **44 zettabytes.** "

Learning from data

2020 This Is What Happens In An Internet Minute



Volume of data/information created, captured, copied, and consumed worldwide from 2010 to 2024 (in zettabytes).



Source: Statista, 2021

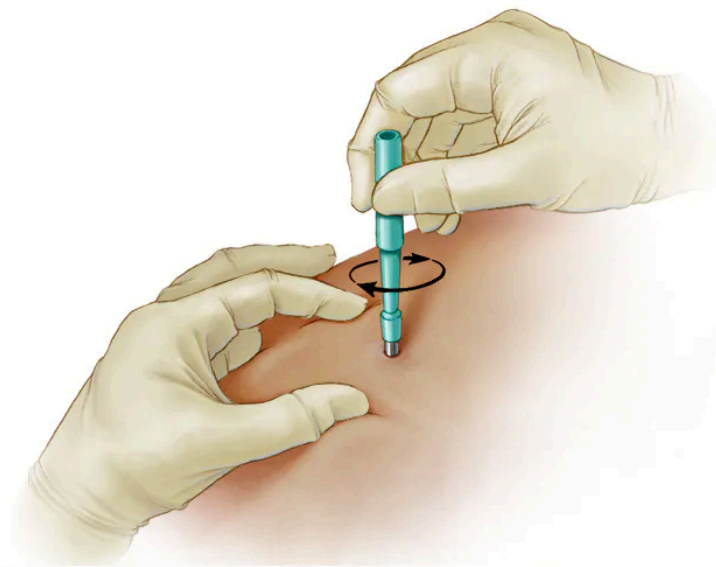


Machine Learning

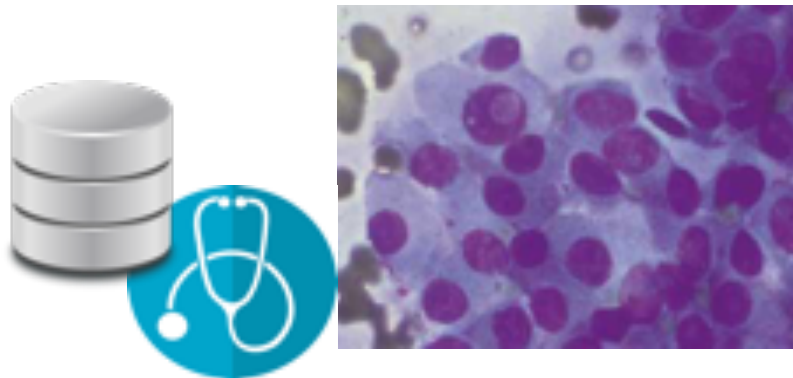
- **Supervised Learning** → Learning to make predictions
Based on a dataset with examples of what is the “right” decision/action in each case
If the attributes of the patient are similar to X, then the right decision is to say they will probably have cancer
- **Unsupervised Learning** → Learning to describe data and find patterns
The customers of my store seem to behave in one of three different ways, based on their purchasing patterns

Supervised Learning: Terminology

Breast Cancer Diagnosis



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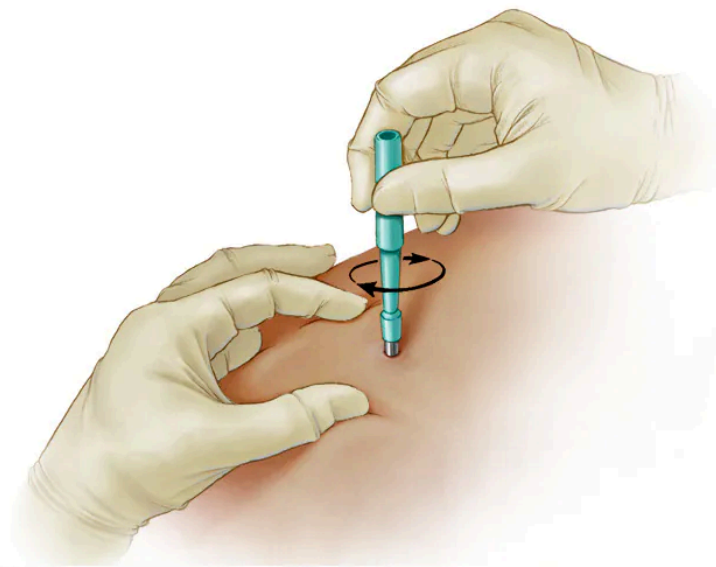


Dataset (WDBC)

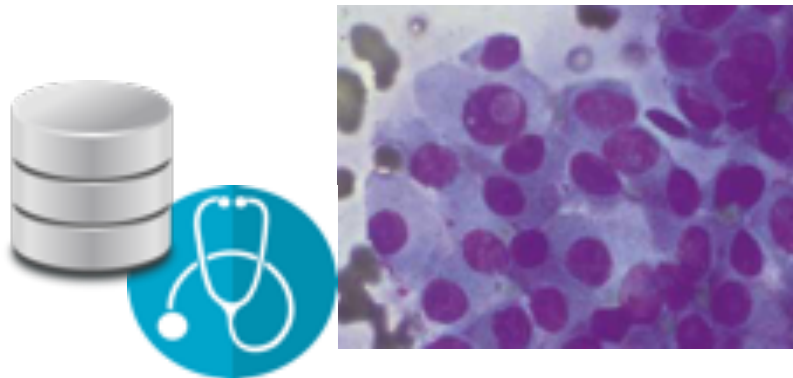
Radius	Texture	Perimeter
14	23	94
15	28	97
15	20	95
16	21	108
20	22	130
15	14	102
9	21	55
9	14	59
9	21	60
11	19	72
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Supervised Learning: Terminology

Breast Cancer Diagnosis



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Dataset (WDBC)

Radius	Texture	Perimeter
14	23	94
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9	21	55
9	14	59
9	21	60
11	19	72
9	17	59
13	16	81

Instance

One example in the training set
Composed of a set of *attributes* or *features*

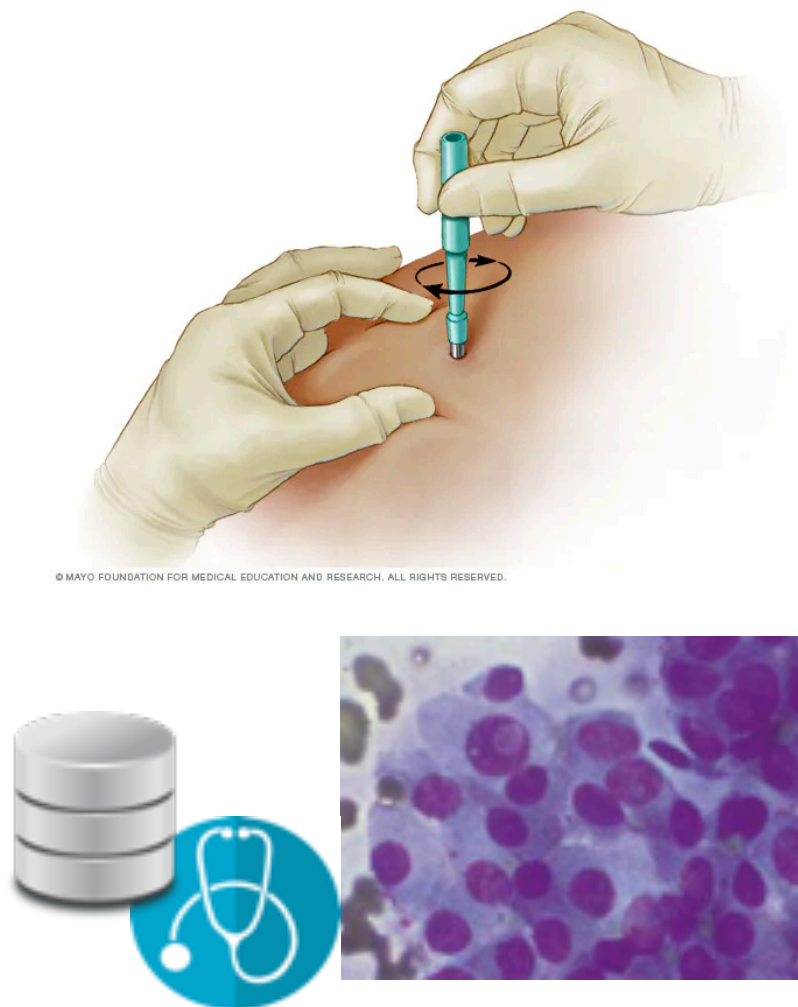
Supervised Learning: Terminology

Breast Cancer Diagnosis

Attribute / Feature

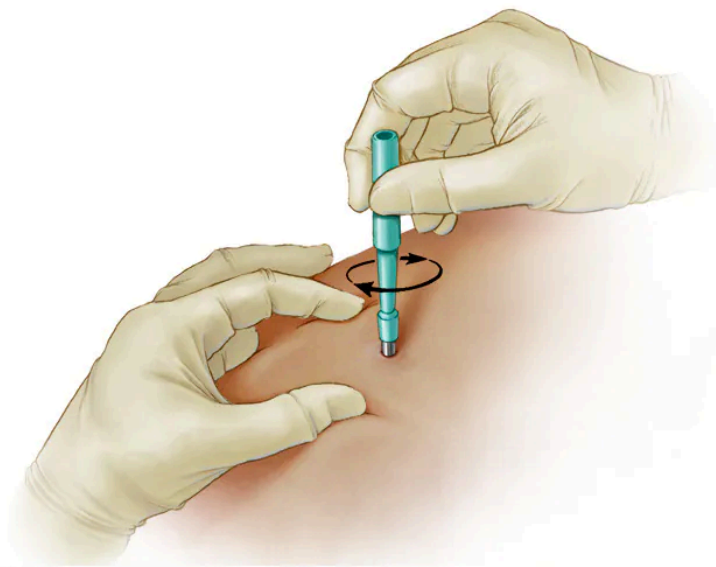
Radius	Texture	Perimeter
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15	28	97
15	20	95
16	21	108
20	22	130
15	14	102
9	21	55
9	14	59
9	21	60
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A property or characteristic of the instance/example

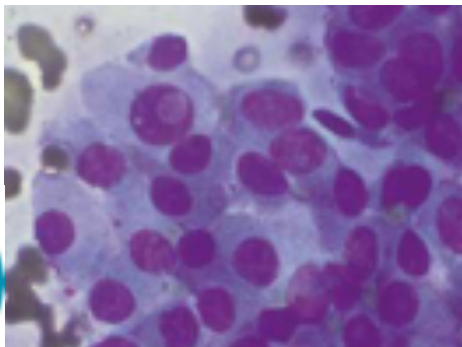


Supervised Learning: Terminology

Breast Cancer Diagnosis



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Dataset (WDBC)

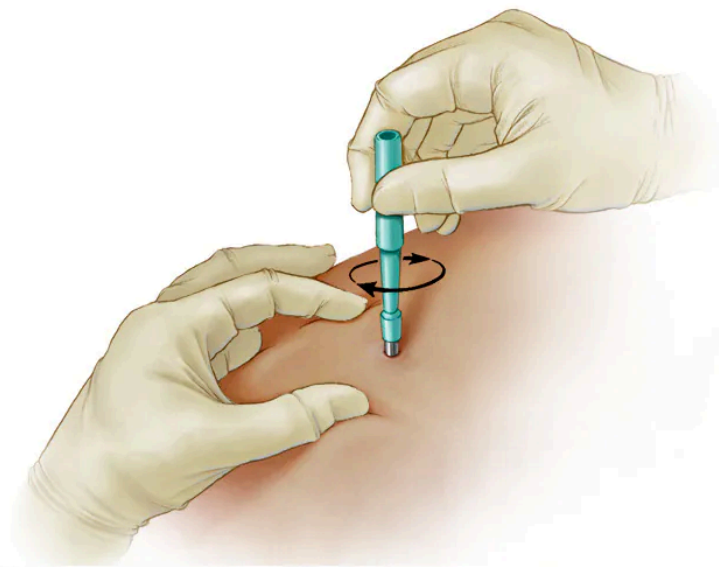
Radius	Texture	Perimeter
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16	21	108
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9	14	59
9	21	60
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13	16	81

x_1

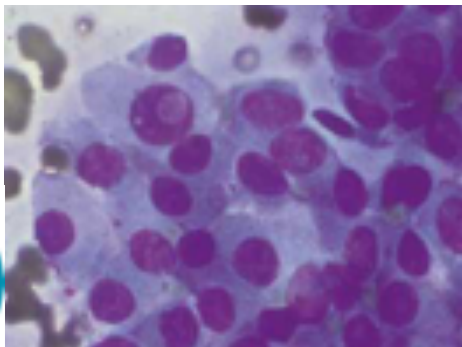
x_4

Supervised Learning: Terminology

Breast Cancer Diagnosis



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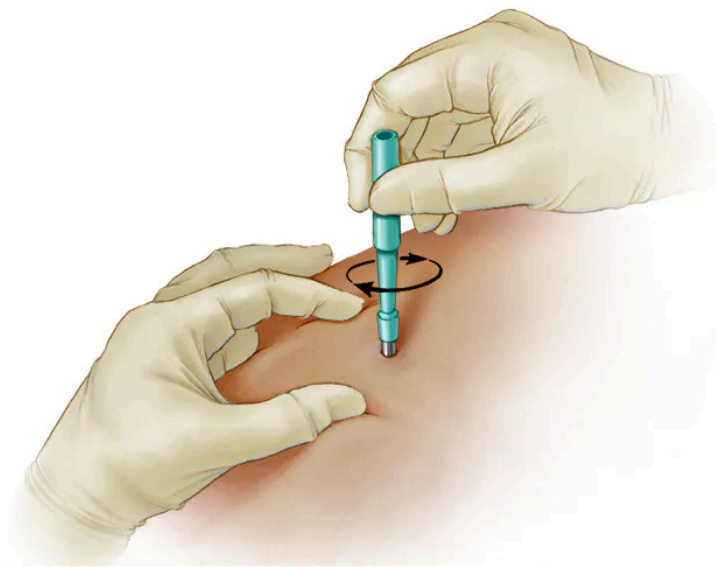
Dataset (WDBC)

Radius	Texture	Perimeter	Diagnosis
14	23	94	M
15	28	97	M
15	20	95	M
16	21	108	M
20	22	130	M
15	14	102	M
9	21	55	B
9	14	59	B
9	21	60	B
11	19	72	B
9	17	59	B
13	16	81	B

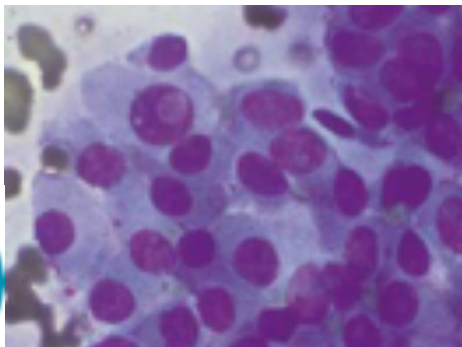
Class / Label

Supervised Learning: Terminology

Breast Cancer Diagnosis



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Dataset (WDBC)

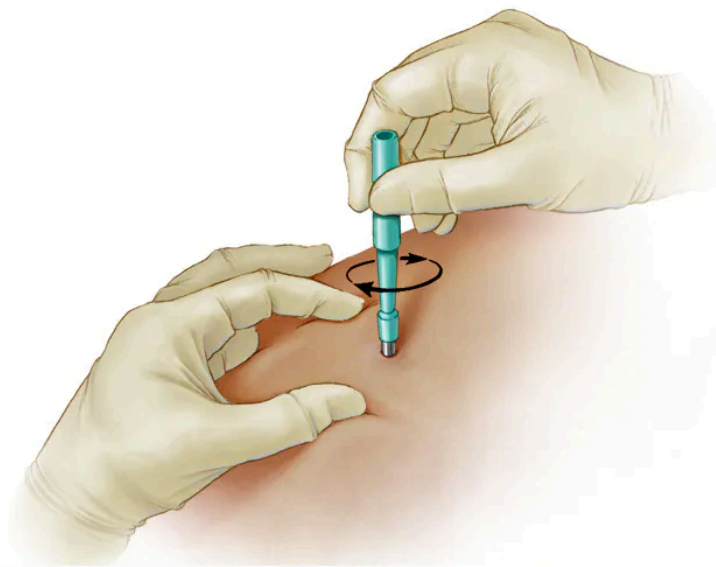
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15	20	95	M
16	21	108	M
20	22	130	M
15	14	102	M
9	21	55	B
9	14	59	B
9	21	60	B
11	19	72	B
9	17	59	B
13	16	81	B

y_1

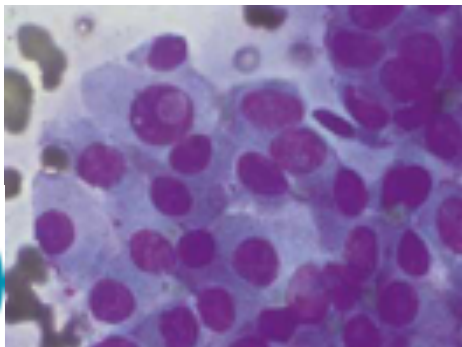
y_4

Supervised Learning: Terminology

Breast Cancer Diagnosis



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Dataset (WDBC)

Radius	Texture	Perimeter	Diagnosis
14	23	94	M
15	28	97	M
15	20	95	M
16	21	108	M
20	22	130	M
15	14	102	M
9	21	55	B
9	14	59	B
9	21	60	B
11	19	72	B
9	17	59	B
13	16	81	B

Training instance #4: $\{x_4, y_4\}$

Supervised Learning: Terminology

Dataset (WDBC)

Radius	Texture	Perimeter	Diagnosis
14	23	94	M
15	28	97	M
15	20	95	M
16	21	108	M
20	22	130	M
15	14	102	M
9	21	55	B
9	14	59	B
9	21	60	B
11	19	72	B
9	17	59	B
13	16	81	B

Class/labels are *categorical features*
Discrete, finite number of possibilities
(either M or B)

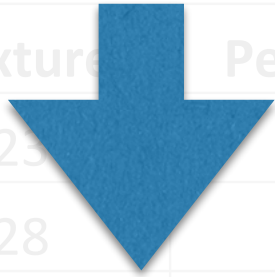


	#friends	funny score	Likes
Post1	40	8	28
Post2	36	10	28
Post3	20	6	16
Post4	56	4	31
Post5	58	0	29
Post6	46	10	33

Values being predicted are *numeric*
Possibility infinite

Supervised Learning: Terminology

Dataset (WDBC)



Radius	Texture	Perimeter	Diagnosis
14	23	94	M
15	28	97	M
15	20	95	M
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9	21	60	B
11	19	72	B
9	17	59	B
13	16	81	B

Classification problems

Class/labels are *categorical features*
Discrete, finite number of possibilities
(either **M** or **B**)



	#friends	funny score	Likes
Post1	40	8	28
Post2	36	10	28
Post3	20	6	16
Post4	56	4	31
Post5	58	0	29
Post6	46	10	33

Regression problems

Values being predicted are *numeric*
Possibility infinite

Learning: Searching Through Candidate Hypotheses

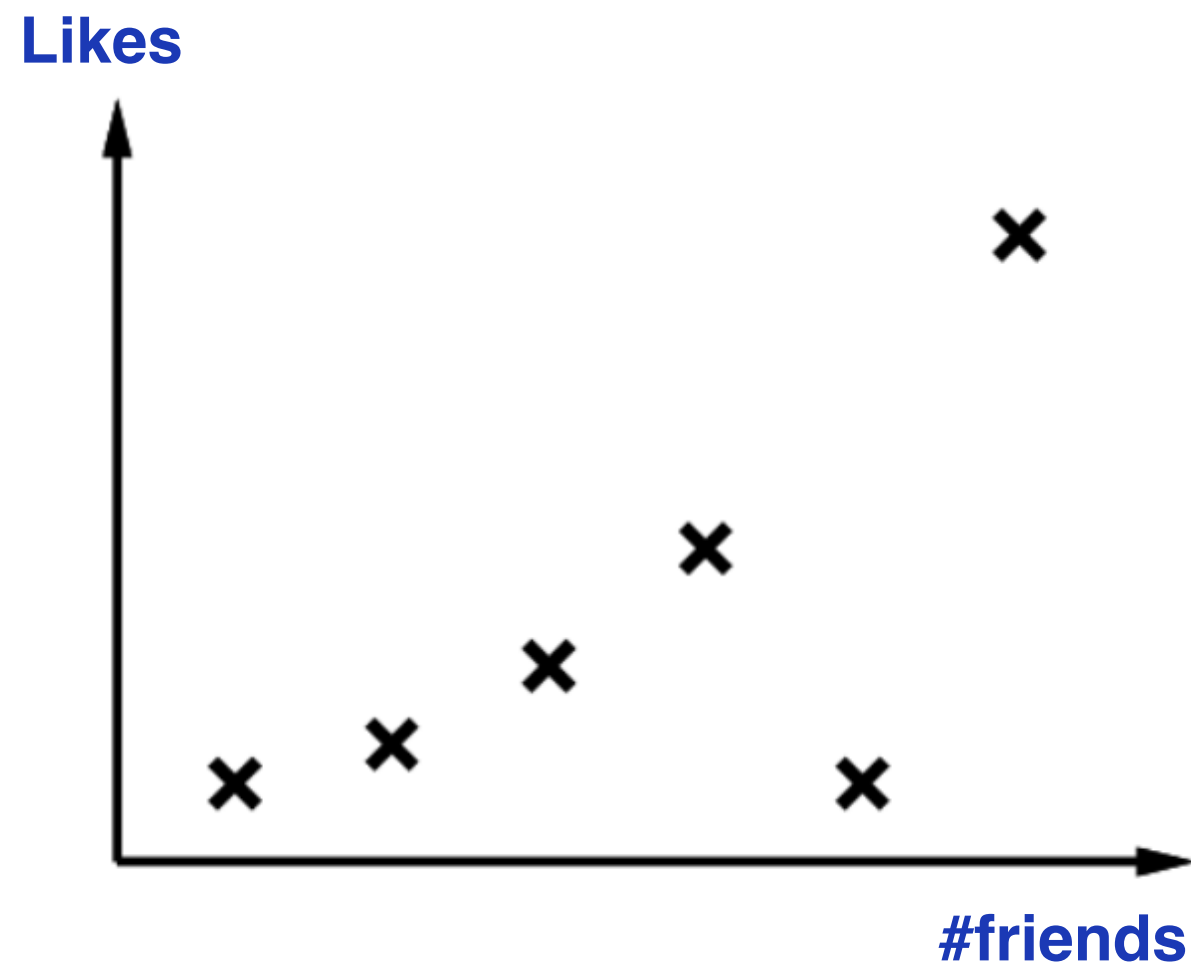


	#friends	funny score	Likes
Post1	40	8	28
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Learning: Searching Through Candidate Hypotheses

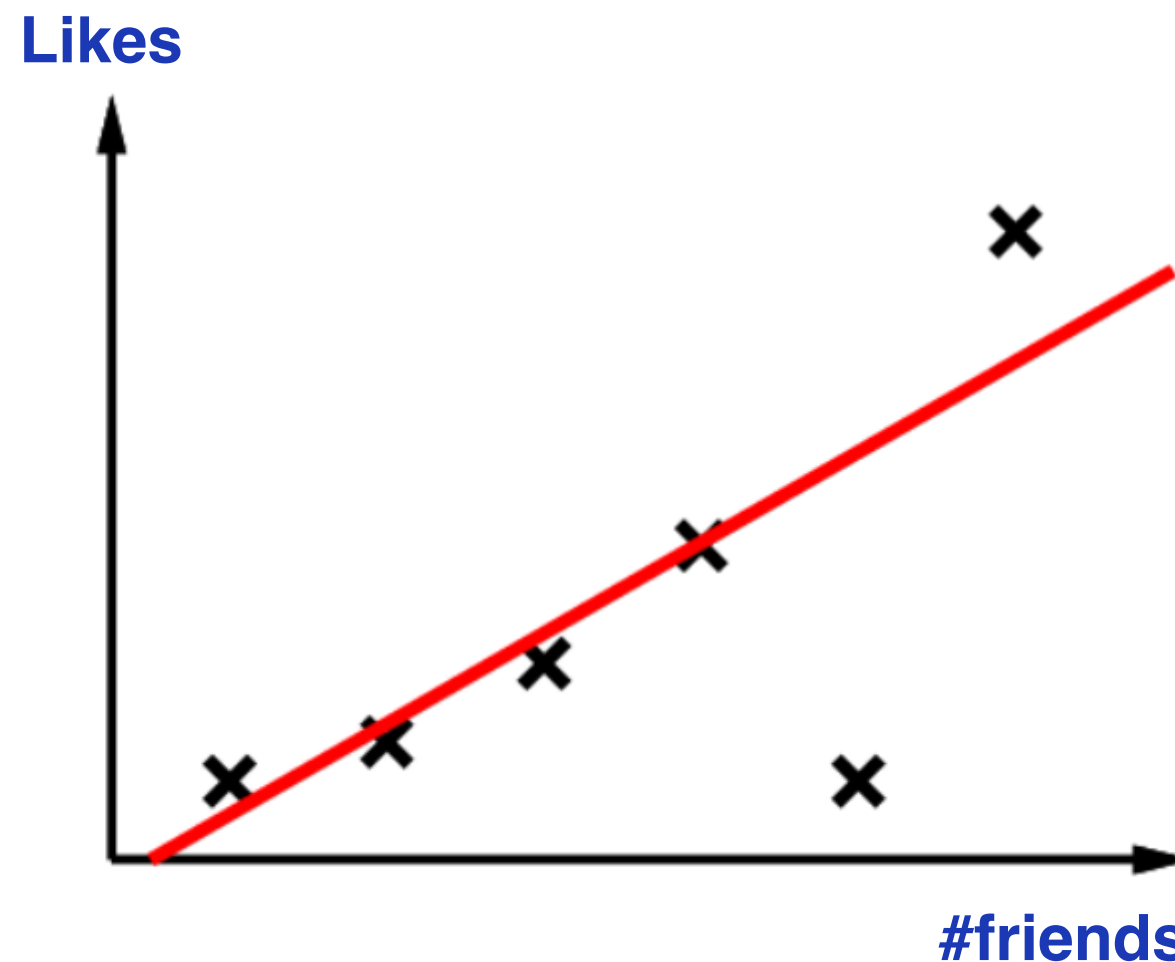


	#friends	Likes
Post1	40	28
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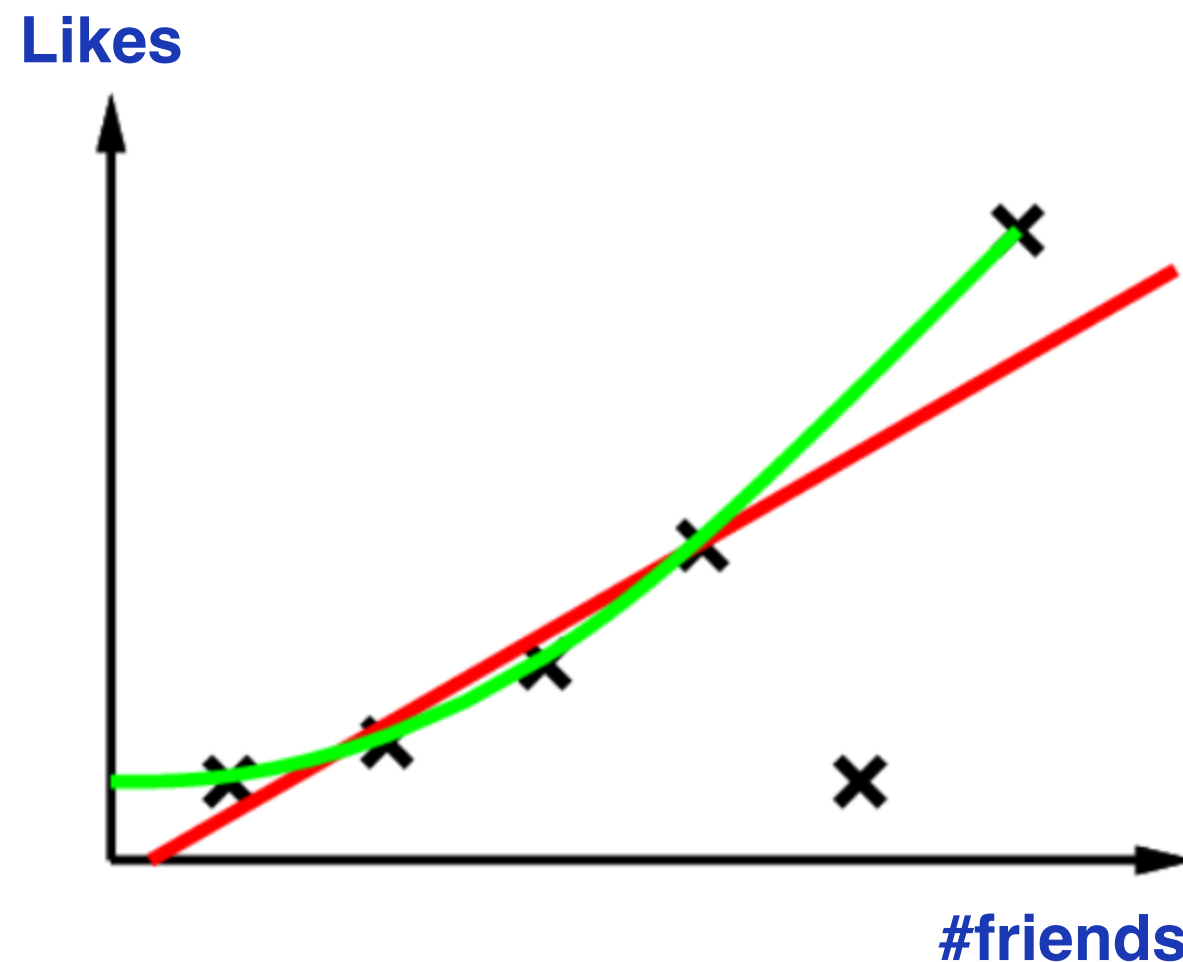
Search through the space of candidate hypotheses
for the one that
"better explains the training data"

Learning: Searching Through Candidate Hypotheses



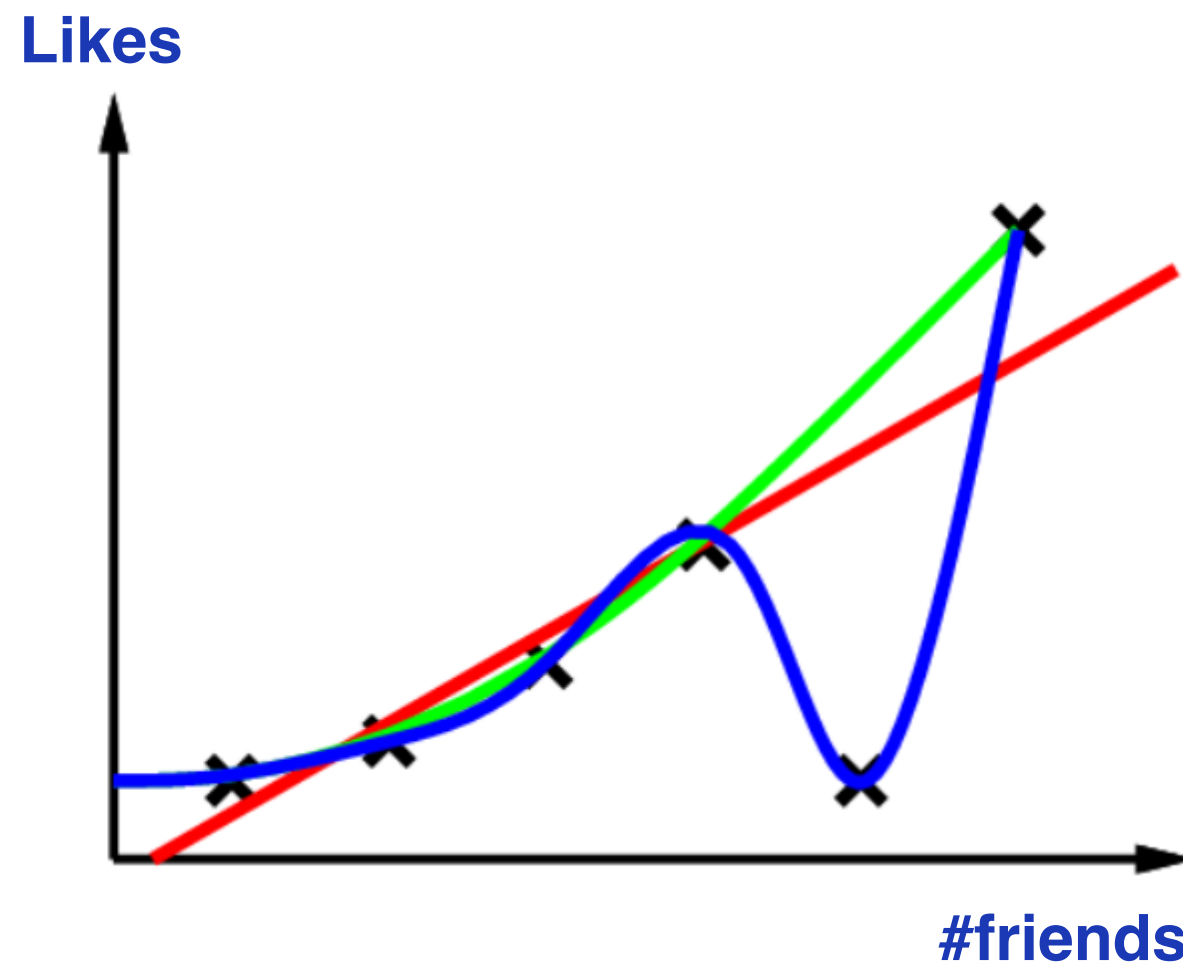
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Learning: Searching Through Candidate Hypotheses



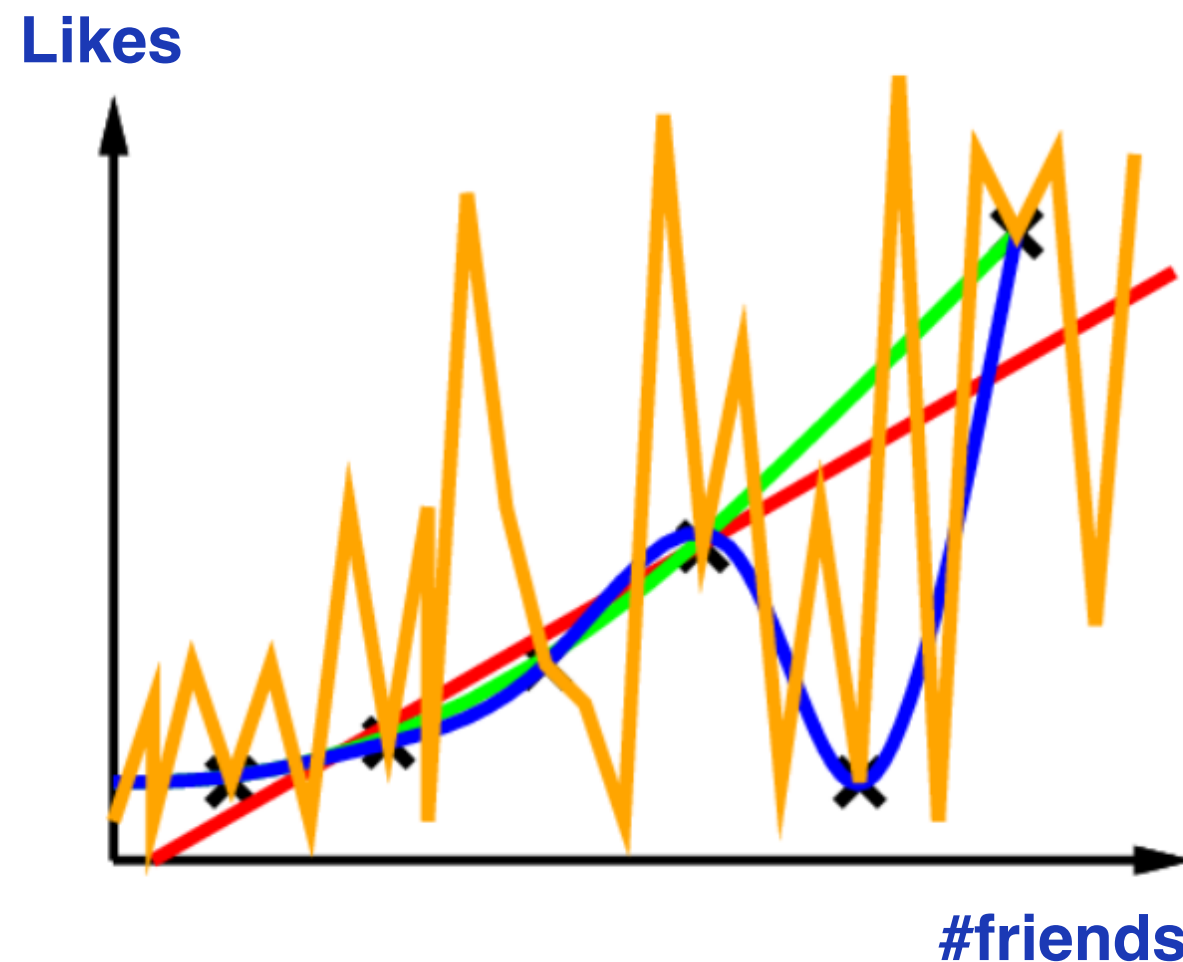
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Learning: Searching Through Candidate Hypotheses



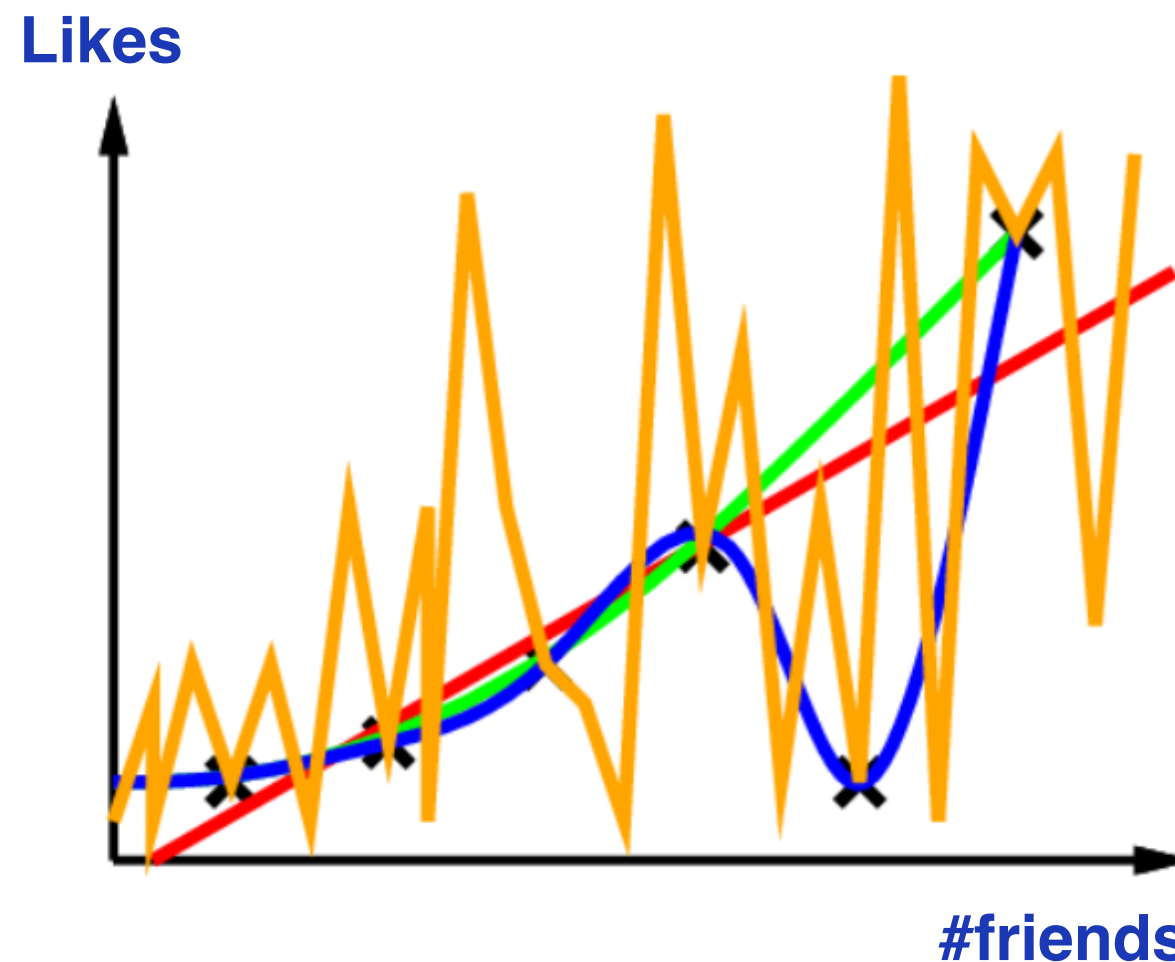
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Learning: Searching Through Candidate Hypotheses



Search through the space of candidate hypotheses
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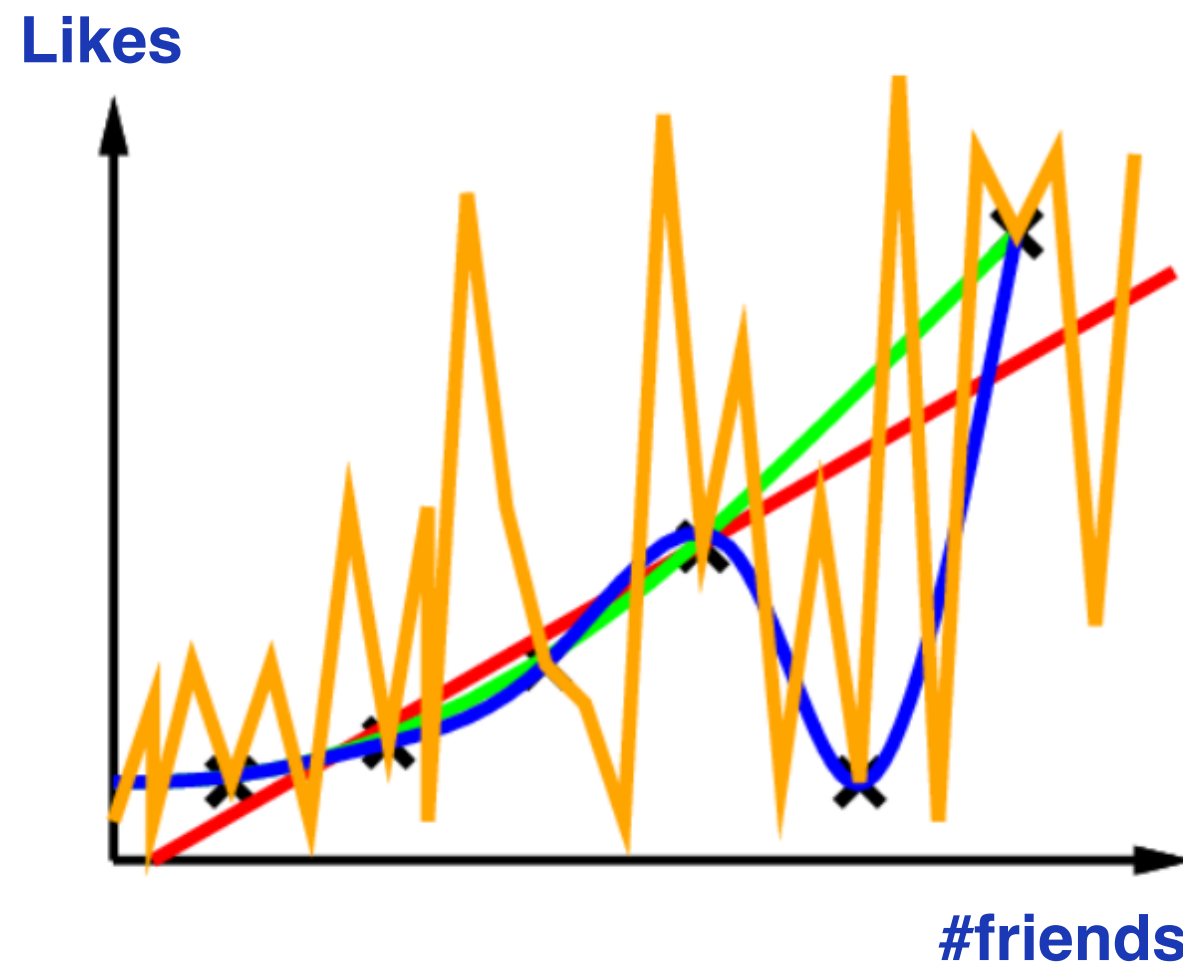
Learning: Searching Through Candidate Hypotheses



Search through the space of candidate hypotheses
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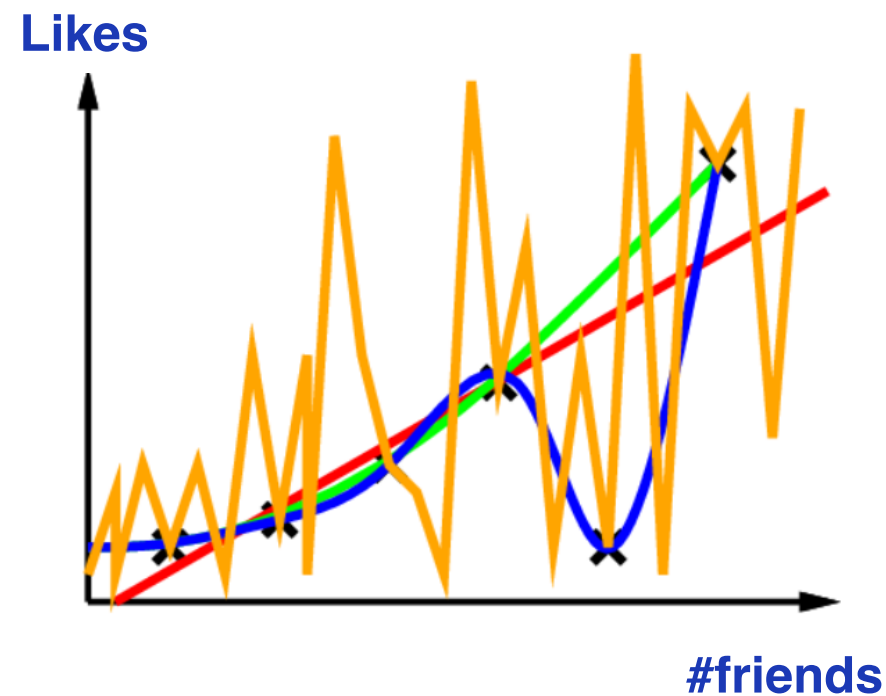
Several hypotheses might be consistent with the training data/examples

Learning: Searching Through Candidate Hypotheses



Inductive Bias

Learning: Searching Through Candidate Hypotheses



Inductive Bias

Different ML algorithms make
(explicit or implicit)
assumptions



Learn different ways
of making predictions

Set of assumptions a ML algorithm uses
to predict outputs of given inputs
that it has not encountered

Classification Algorithms

(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Perimeter	Diagnosis
14	23	94	M
15	28	97	M
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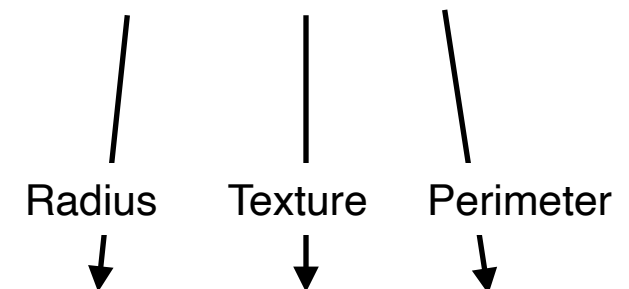
Attributes/Features
(input)

Class/Label
(output)

Training/learning
a prediction model



Attributes of a
new/unknown instance
(input)



Diagnosis

Prediction of the class / label
of the new/unknown instance
(output)

Classification Algorithms

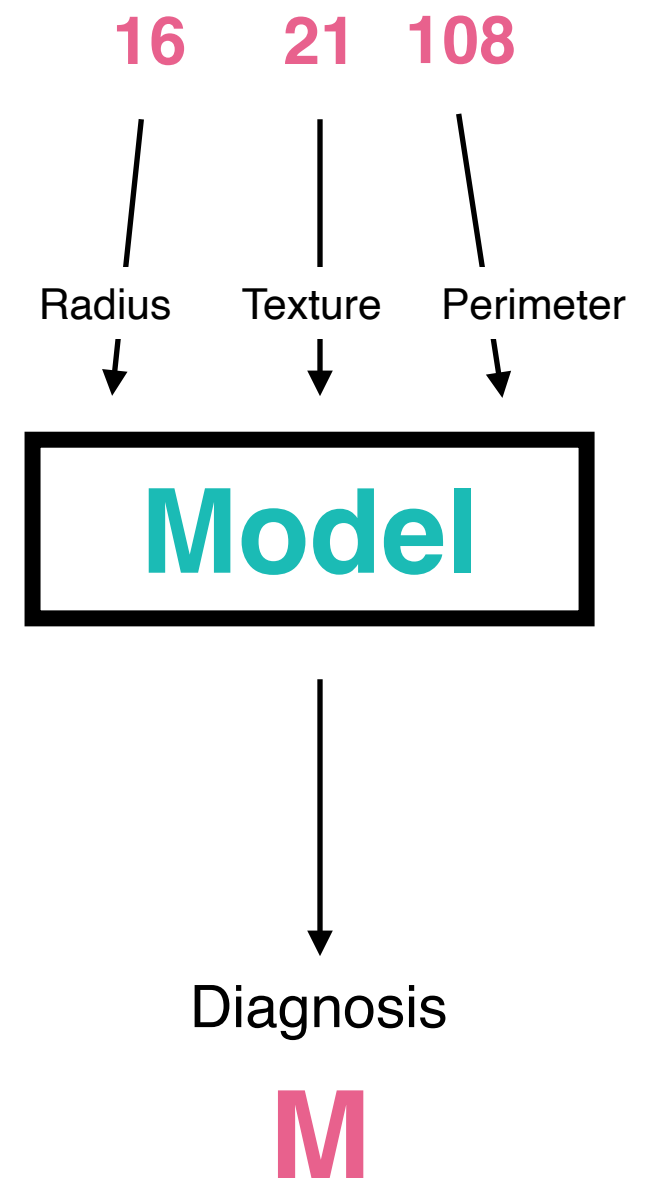
(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

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Attributes/Features (input) Class/Label (output)

Training/learning
a prediction model



Classification Algorithms

(Labeled) examples
provided to the algorithm

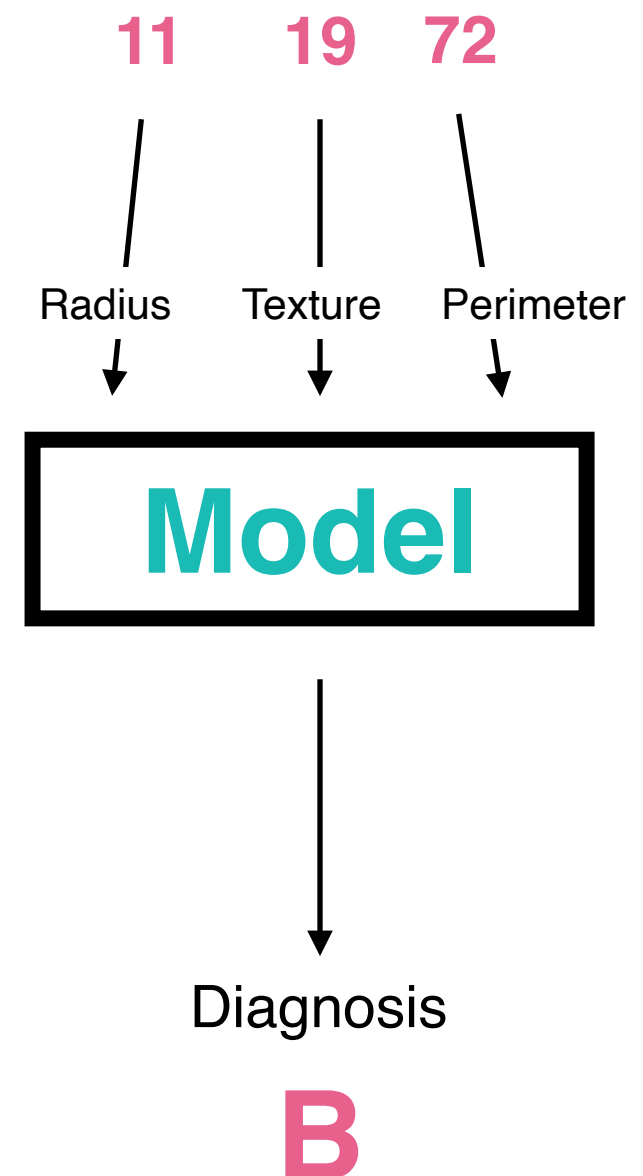
Breast Cancer Diagnosis Dataset (WDBC)

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Attributes/Features
(input)

Class/Label
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Training/learning
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Classification Algorithms

(Labeled) examples
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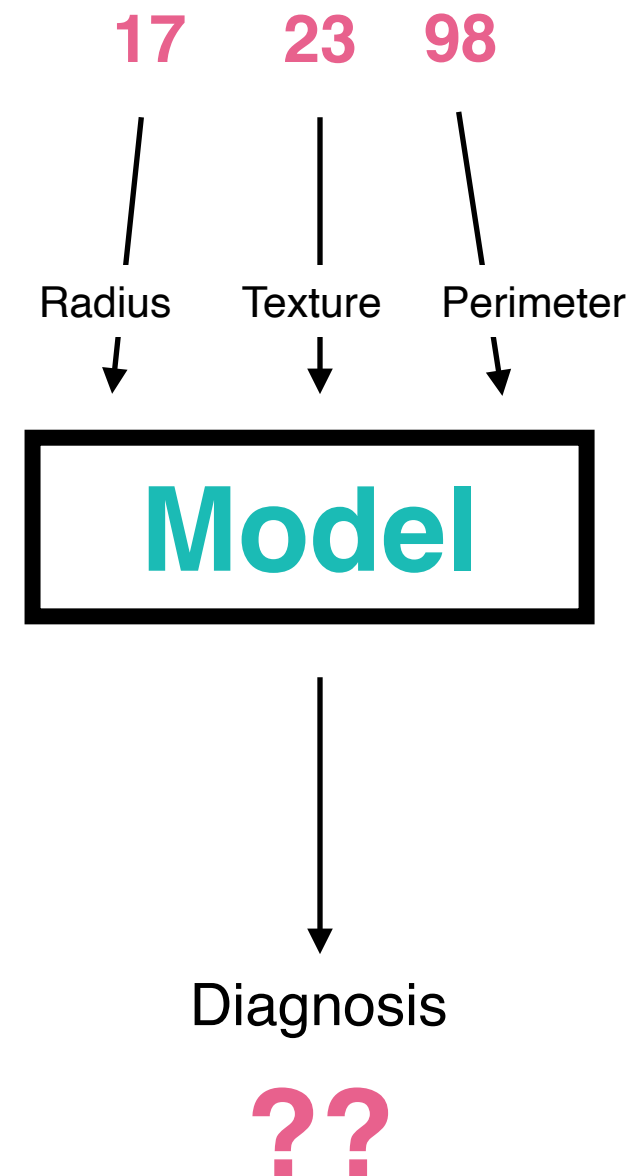
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Attributes/Features (input) Class/Label (output)

17	23	98	??
----	----	----	----

New (unknown/unlabeled) patient



Classification Algorithms

(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Perimeter	Diagnosis
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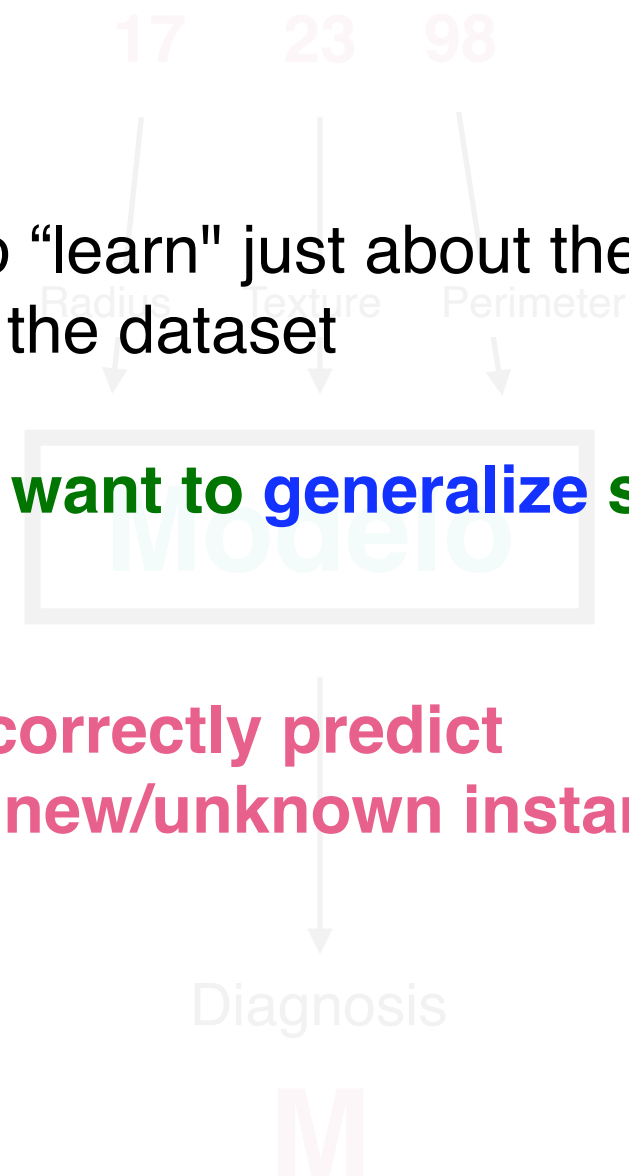
Attributes/Features (input) Class/Label (output)

17	23	98	??
----	----	----	----

New (unknown/unlabeled) patient

Intuitively:

- goal is *not* to “learn” just about the examples in the dataset
- instead, we want to **generalize** such knowledge
- in order to **correctly predict** the label of new/unknown instances

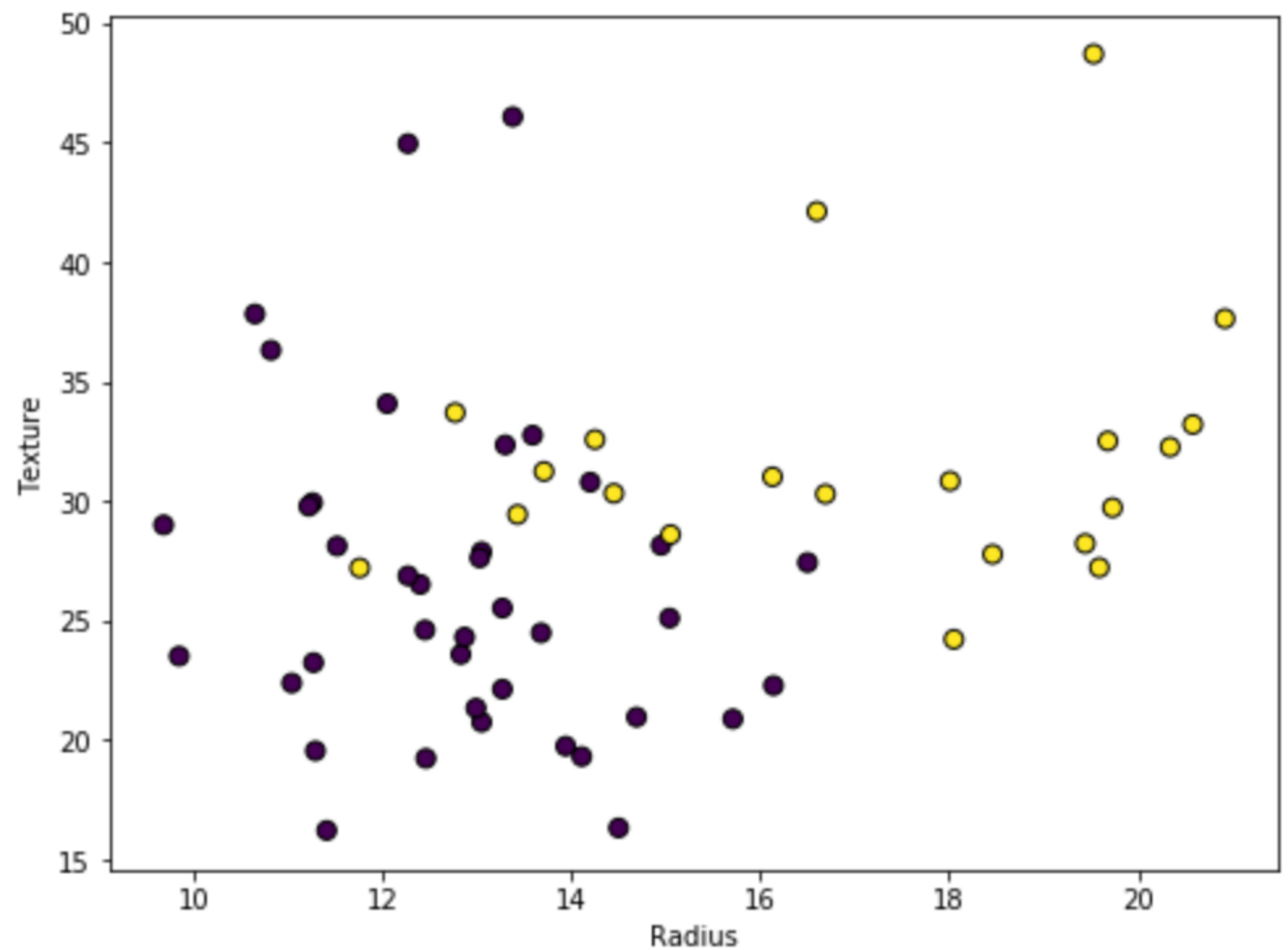


Classification Algorithms

(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Diagnosis
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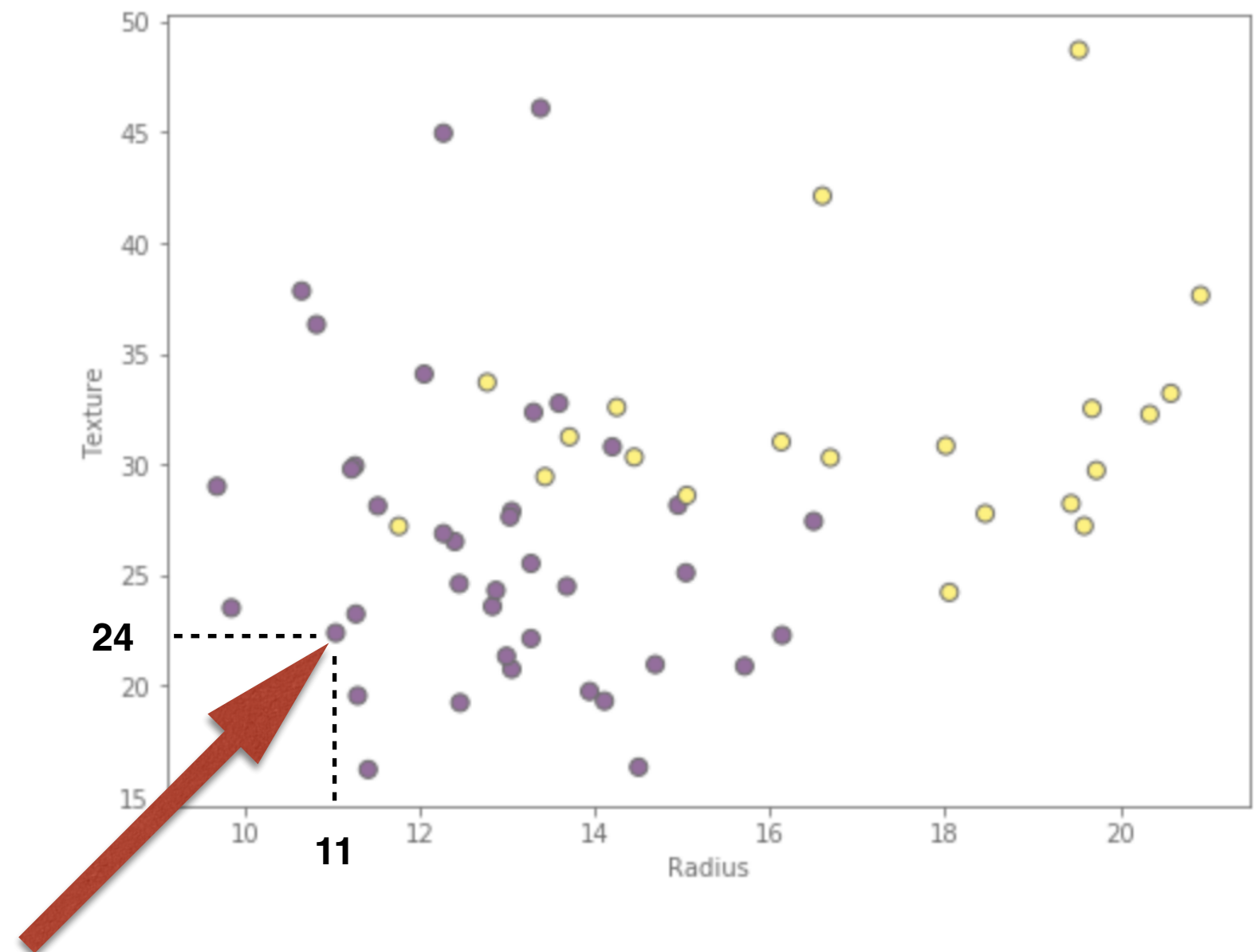


Classification Algorithms

(Labeled) examples
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Breast Cancer Diagnosis Dataset (WDBC)

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13	16	B

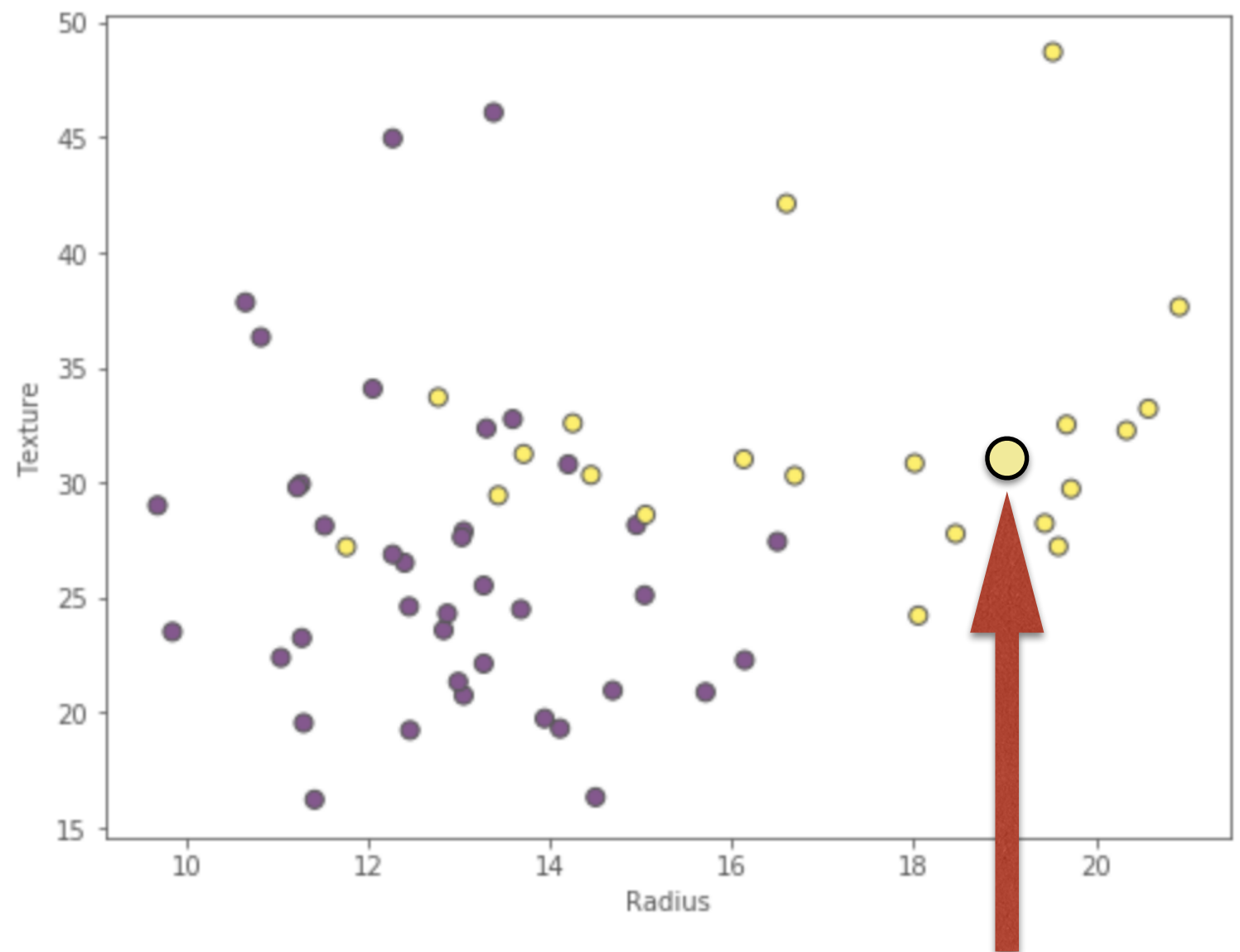


Classification Algorithms

(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Diagnosis
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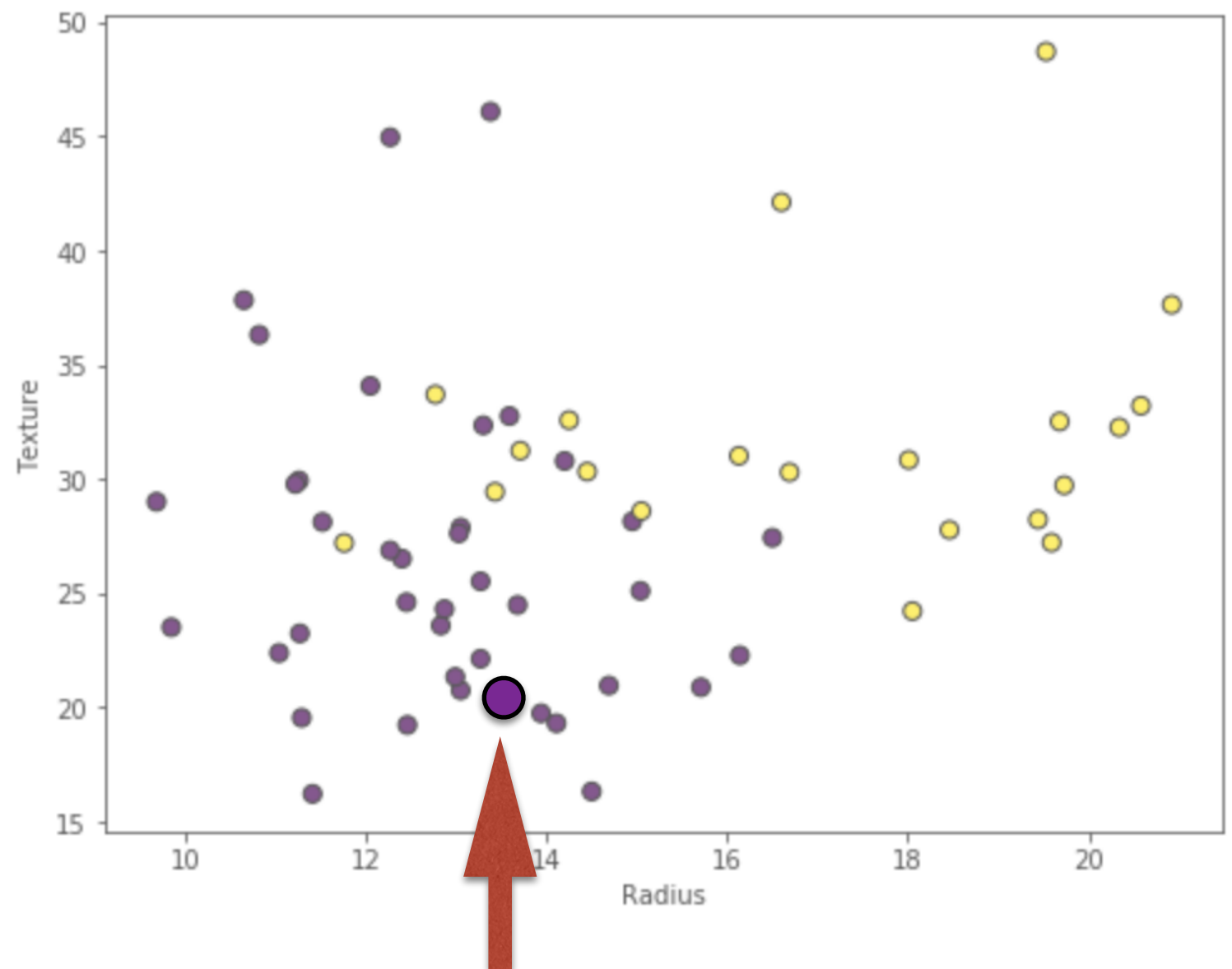
Benign or Malignant?

Classification Algorithms

(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Diagnosis
14	23	M
15	28	M
15	20	M
16	21	M
20	22	M
15	14	M
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13	16	B



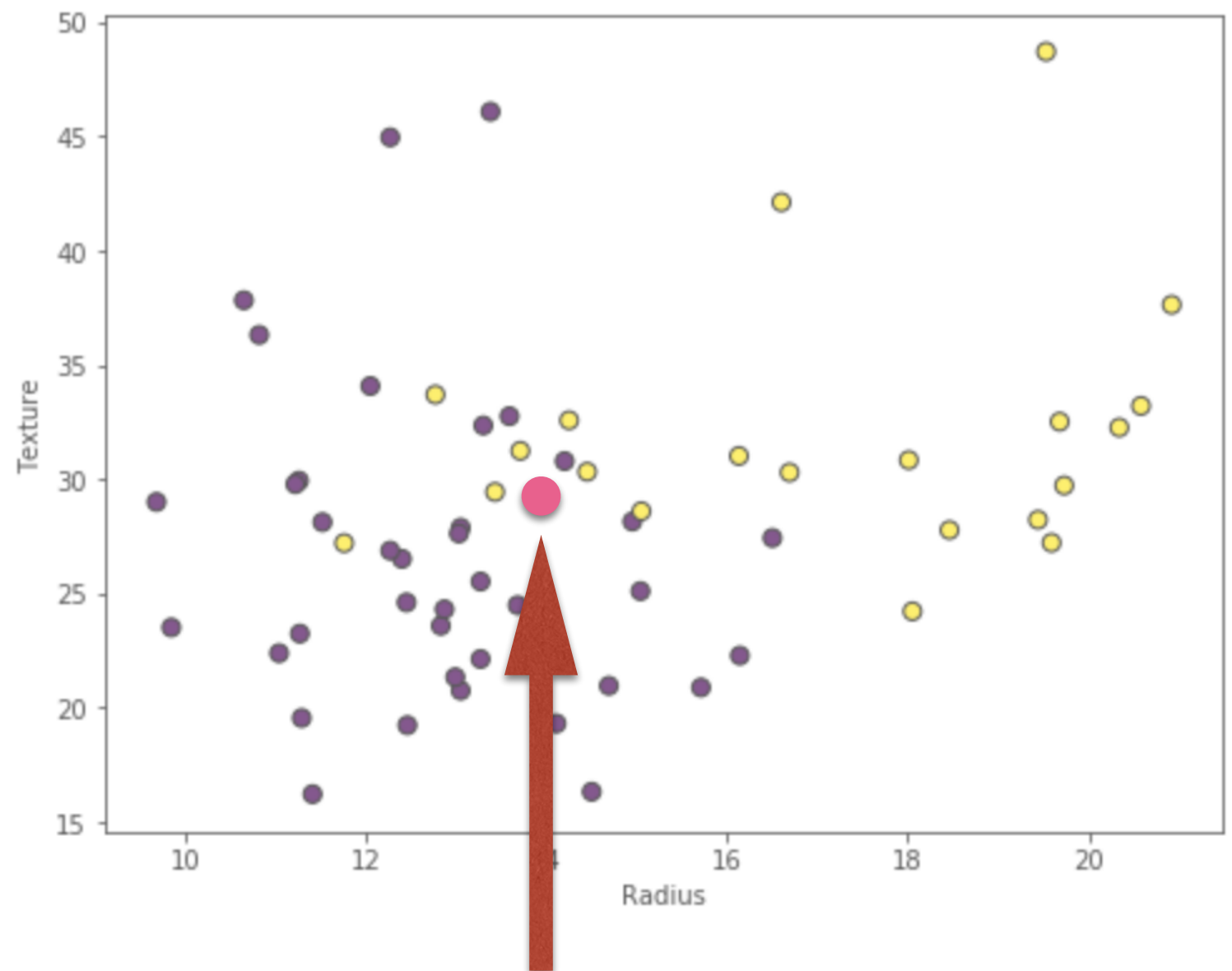
Benign or Malignant?

Classification Algorithms

(Labeled) examples
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9	17	B
13	16	B



Benign or Malignant?

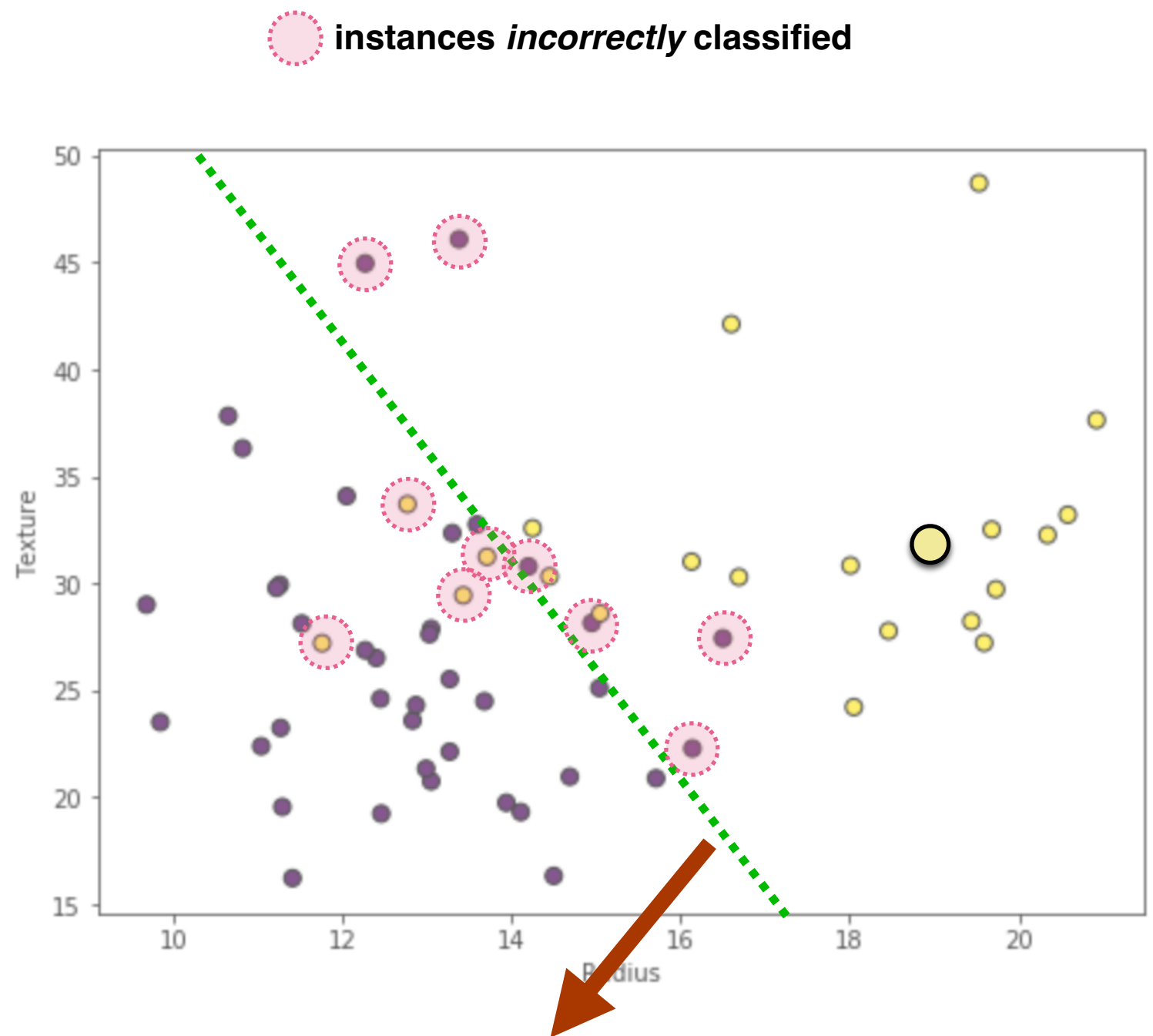
Inconclusive!

Classification Algorithms

(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Diagnosis
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15	28	M
15	20	M
16	21	M
20	22	M
15	14	M
9	21	B
9	14	B
9	21	B
11	24	B
9	17	B
13	16	B



Decision boundary (linear)

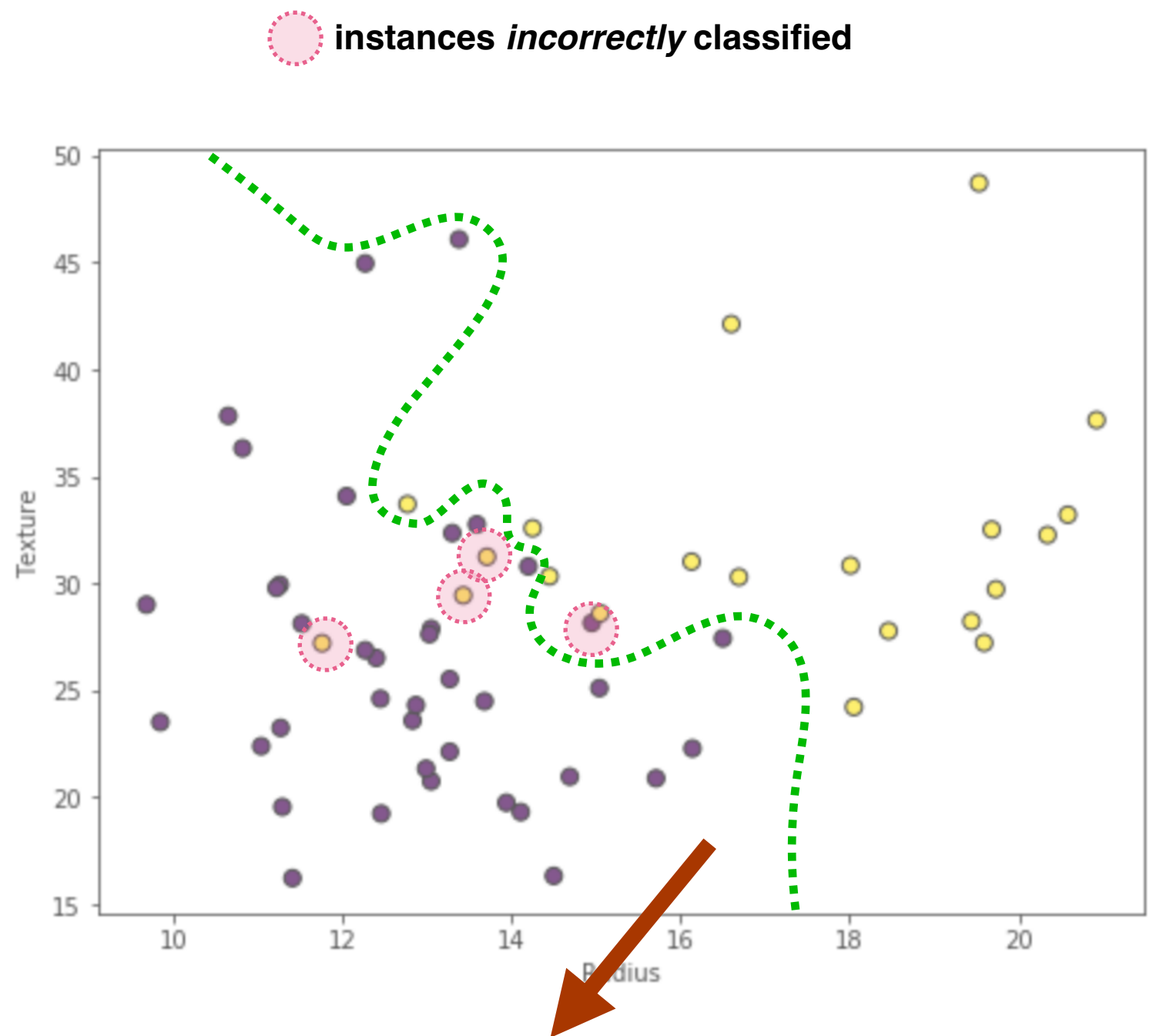
10 classification errors

Classification Algorithms

(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Diagnosis
14	23	M
15	28	M
15	20	M
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9	17	B
13	16	B



Classification Algorithms

(Labeled) examples
provided to the algorithm

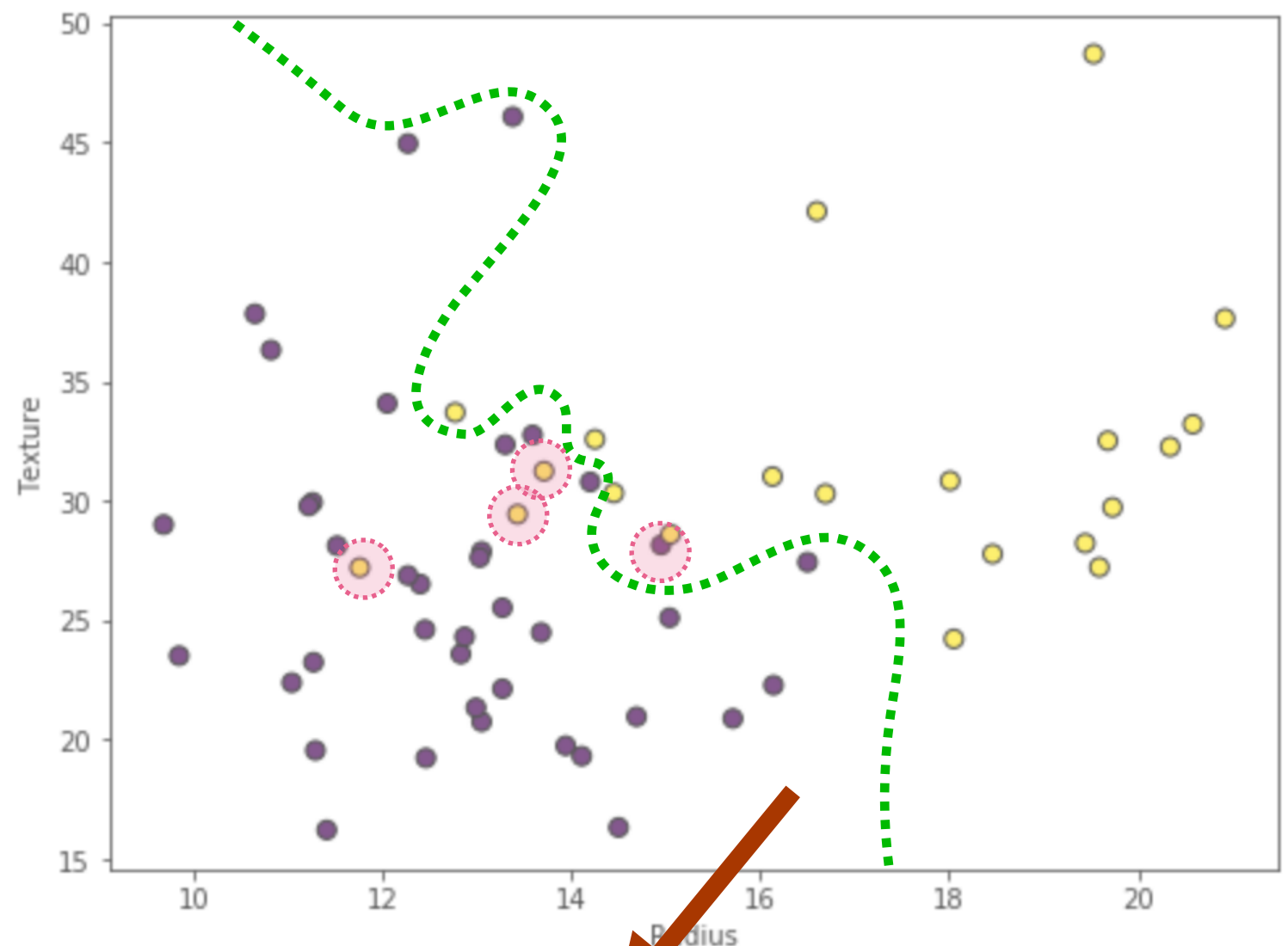
Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Diagnosis
14	23	M
15	28	M
15	20	M
16	21	M
20	22	M
15	14	M
9	21	B
9	14	B
9	21	B
11	24	B
9	17	B
13	16	B

Complex decision boundaries
→ more likely to correctly classify
known instances (examples in dataset)

Is this good, necessarily?

 instances *incorrectly* classified



Decision boundary (non-linear)
4 classification errors

Classification Algorithms

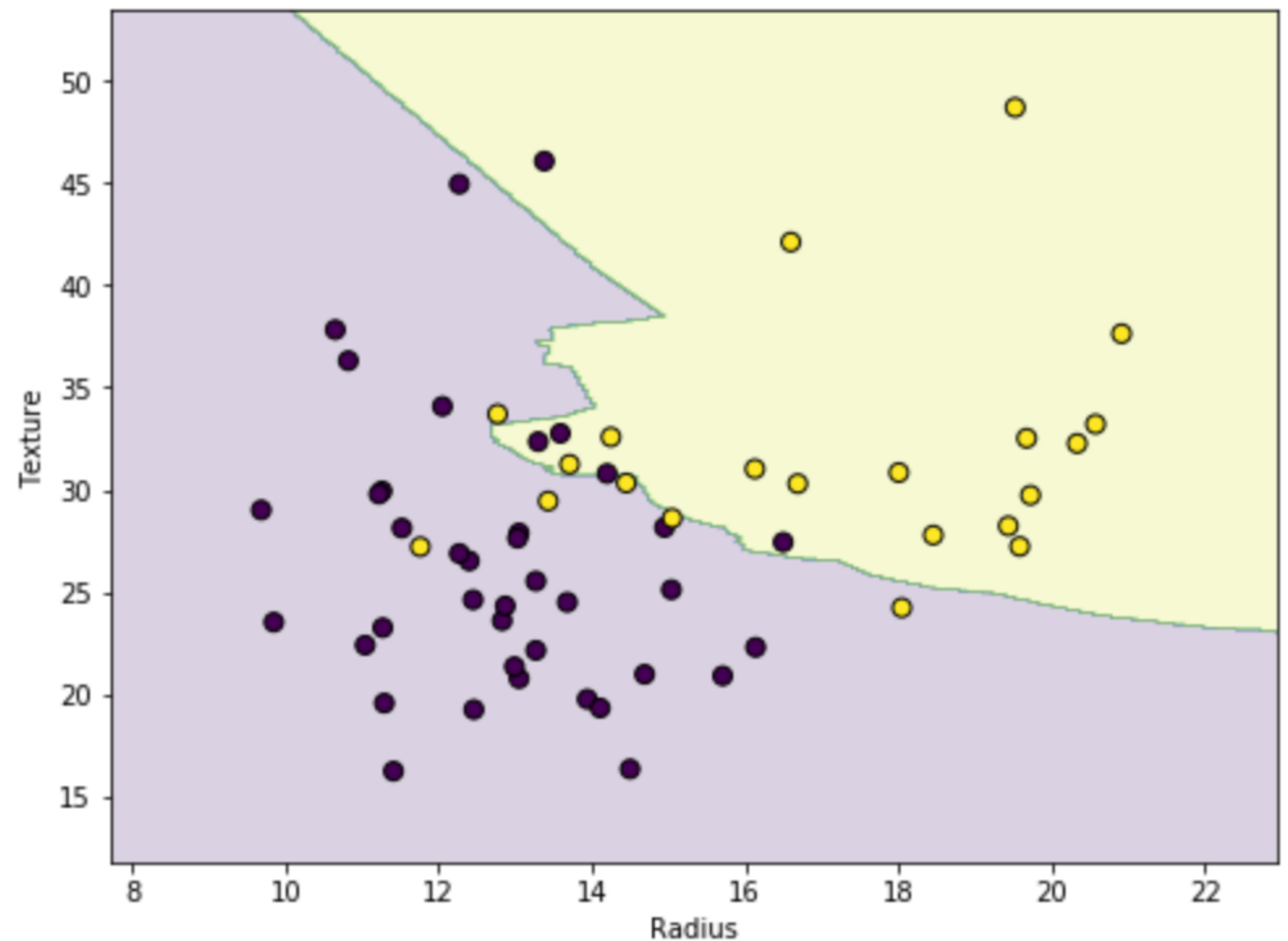
(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Diagnosis
14	23	M
15	28	M
15	20	M
16	21	M
20	22	M
15	14	M
9	21	B
9	14	B
9	21	B
11	24	B
9	17	B
13	16	B

Complex decision boundaries
→ more likely to correctly classify
known instances (examples in dataset)

Is this good, necessarily?



Simple decision boundary
(9 classification errors over known instances)

Classification Algorithms

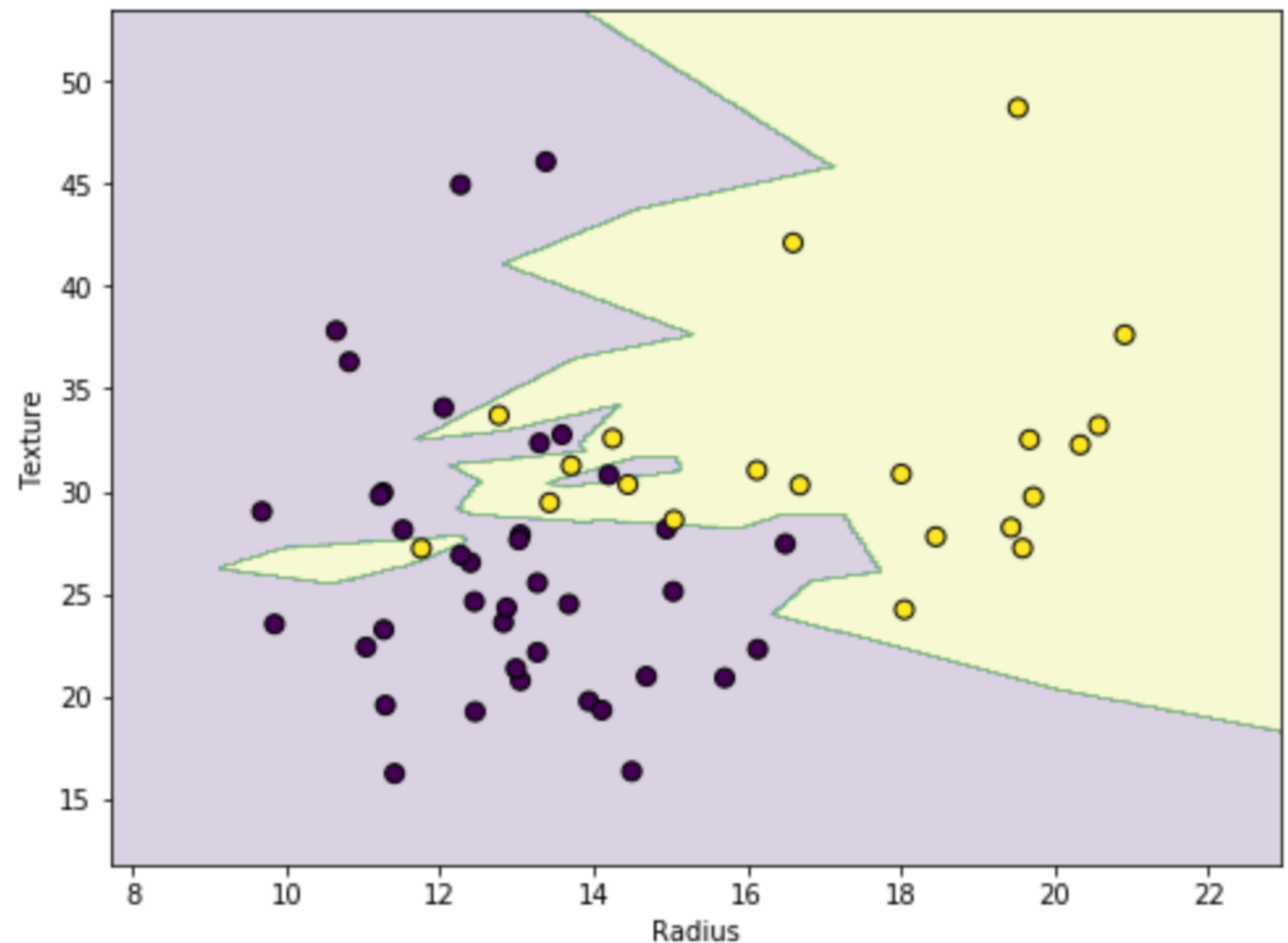
(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Diagnosis
14	23	M
15	28	M
15	20	M
16	21	M
20	22	M
15	14	M
9	21	B
9	14	B
9	21	B
11	24	B
9	17	B
13	16	B

Complex decision boundaries
→ more likely to correctly classify
known instances (examples in dataset)

Is this good, necessarily?



Complex decision boundary
(0 classification errors over known instances!)

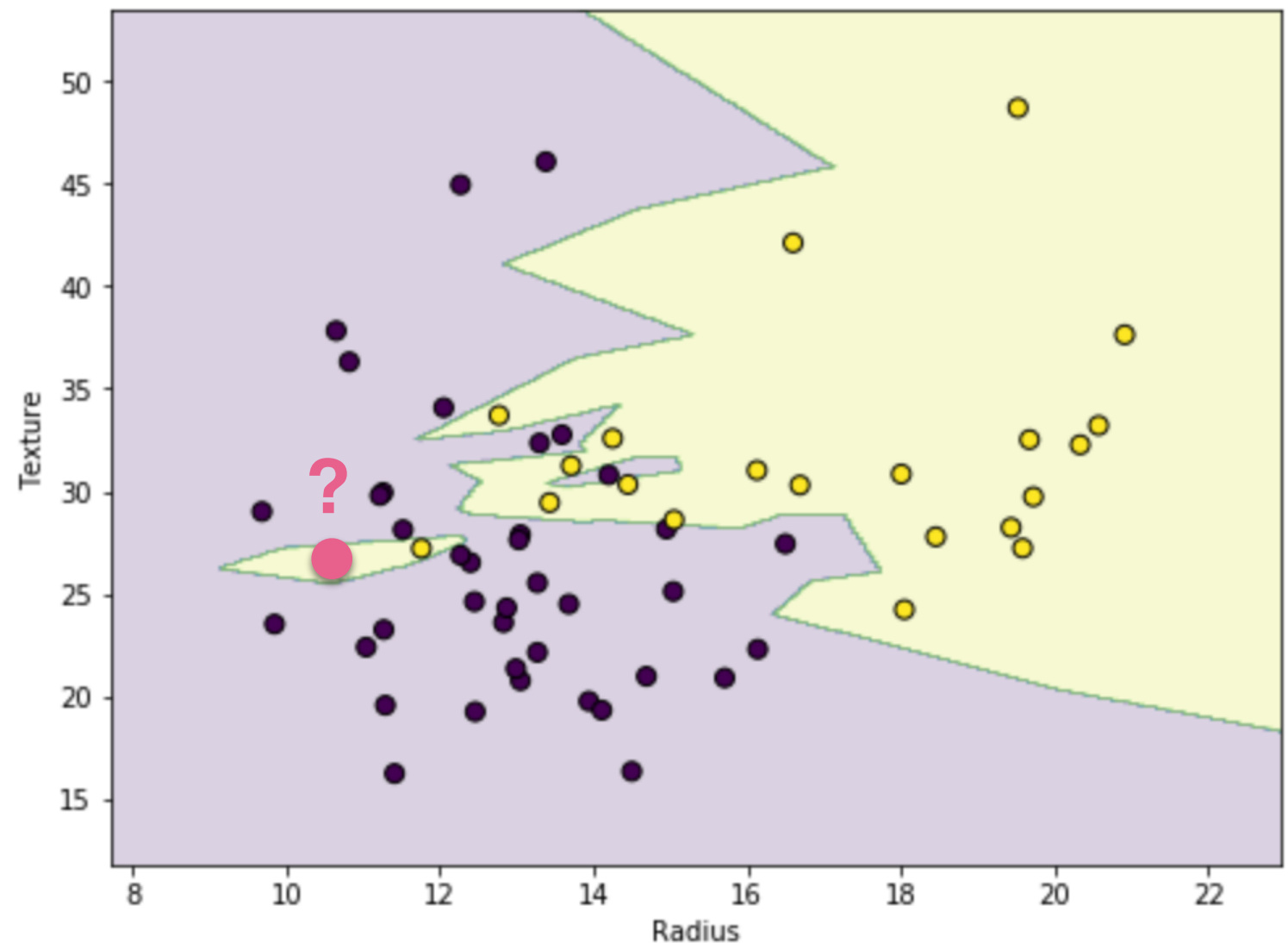
Classification Algorithms

(Labeled) examples
provided to the algorithm

Breast Cancer Diagnosis Dataset (WDBC)

Radius	Texture	Diagnosis
14	23	M
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Classification Algorithms

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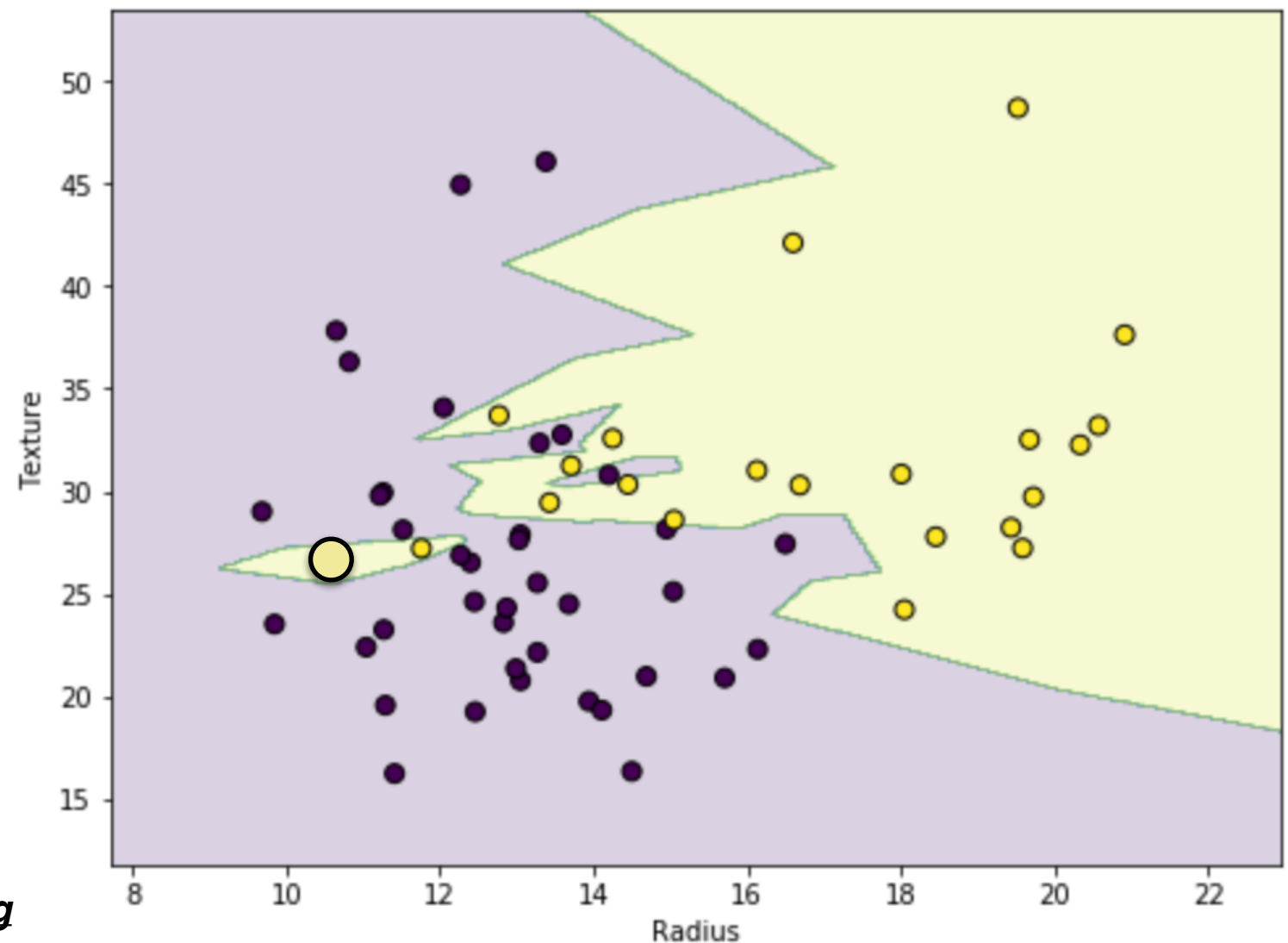
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9	17	B
13	16	B

Probably not the right prediction!

Complex decision boundaries may be overfitting
to the particular examples given to the algorithm

Would probably fail if used
to classify new/unknown instances



Classification Algorithms

(Labeled) examples
provided to the algorithm

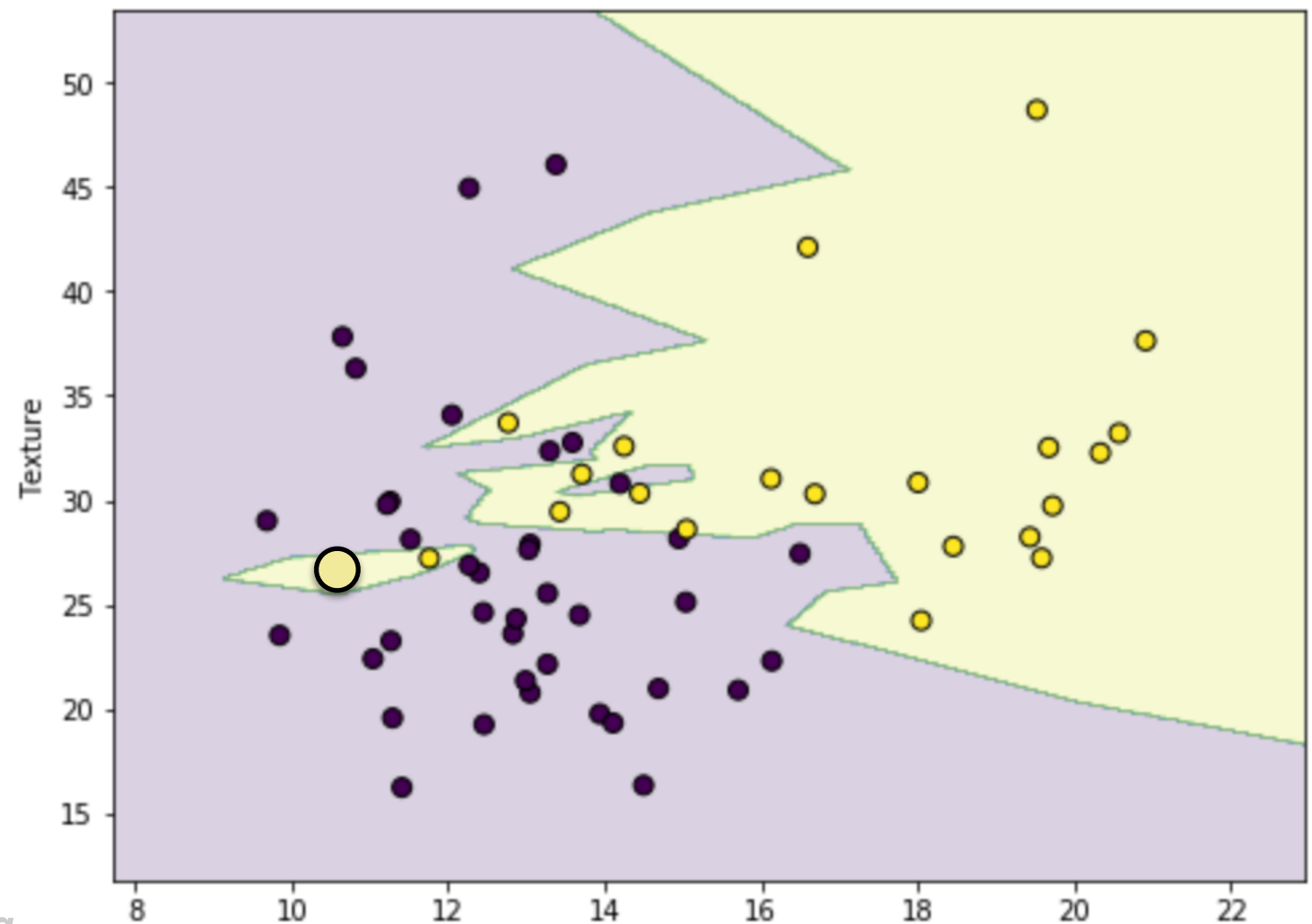
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11	24	B
9	17	B
13	16	B

Probably not the right prediction!

Complex decision boundaries may be *overfitting*
to the particular examples given to the algorithm

Would probably fail if used
to classify new/unknown instances



- Remember:
 - the goal is *not* to “memorize” the dataset
 - but to generalize knowledge to new instances/cases

Instance-Based Classification:

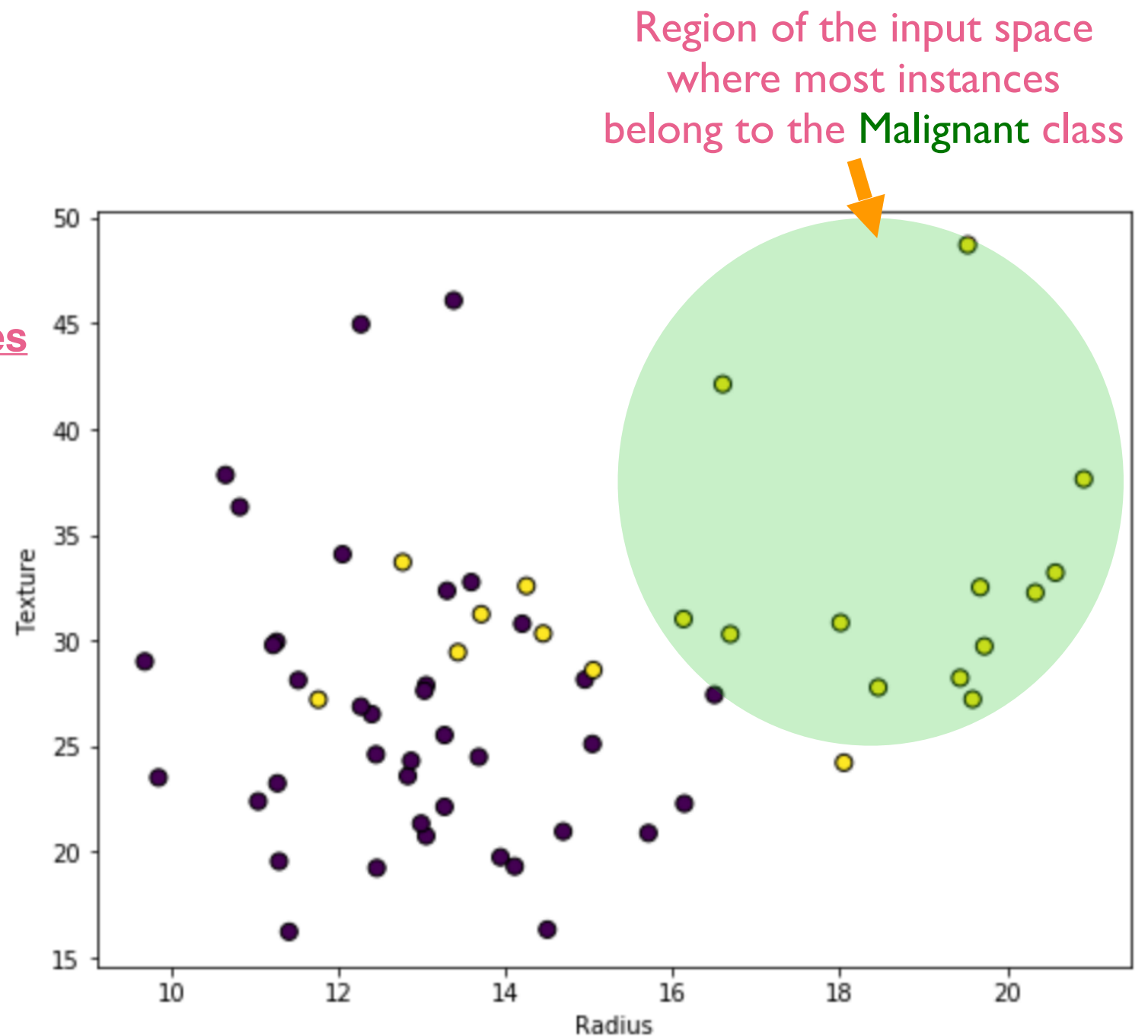
k-nearest neighbors (*k*-NN) algorithm

The k -nearest neighbors (k -NN) algorithm

- Instances belonging to a same class are likely to be near in the space of input attributes
- Instances belonging to different classes are likely to be far away in the space of input attributes

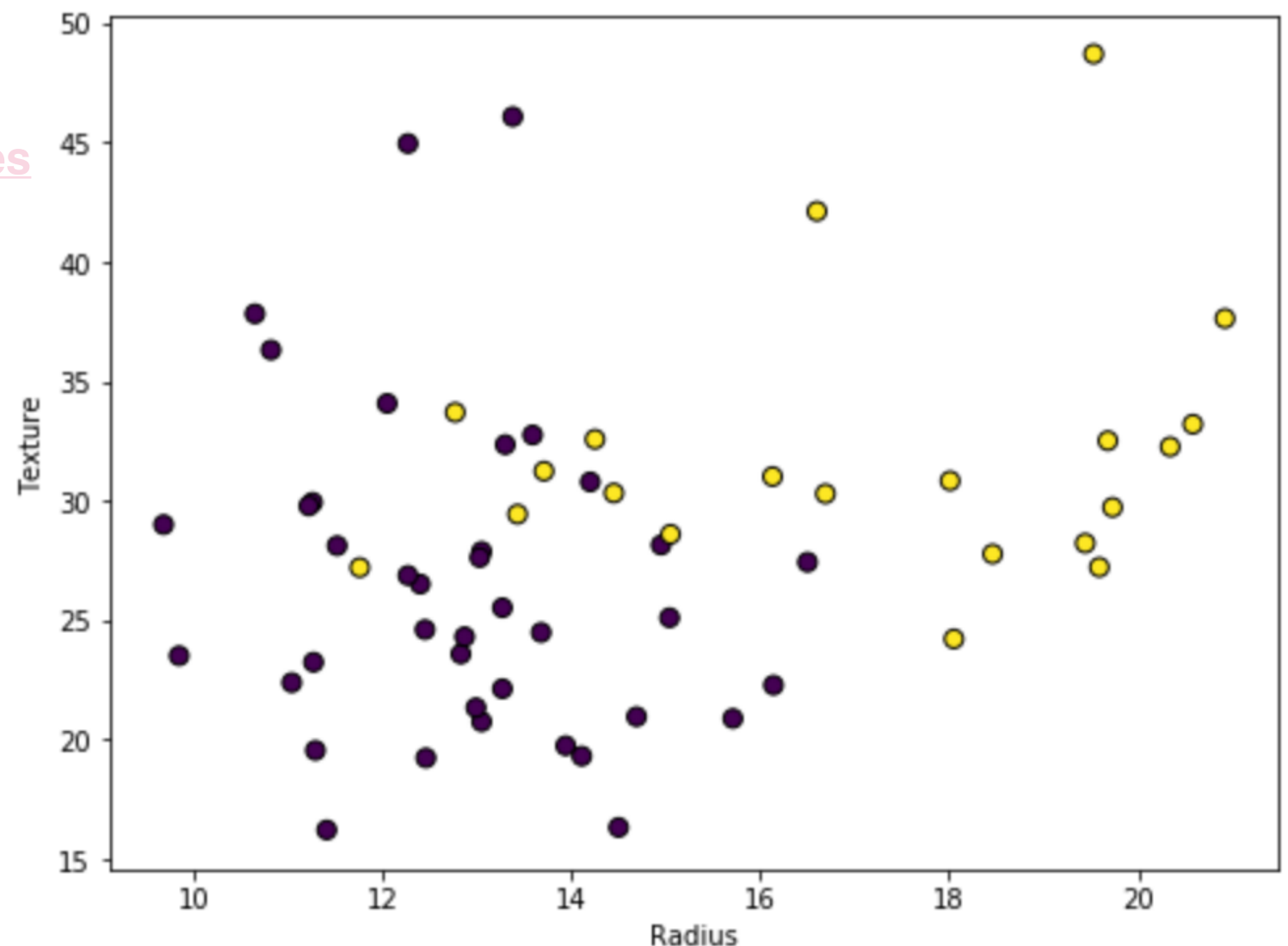


Inductive bias
of k -NN



The k -nearest neighbors (k -NN) algorithm

- Instances belonging to a same class are likely to be near in the space of input attributes
- Instances belonging to different classes are likely to be far away in the space of input attributes
- To classify a new instance (with unknown class/label) its k nearest instances vote
 - Based, e.g., on Euclidean distance



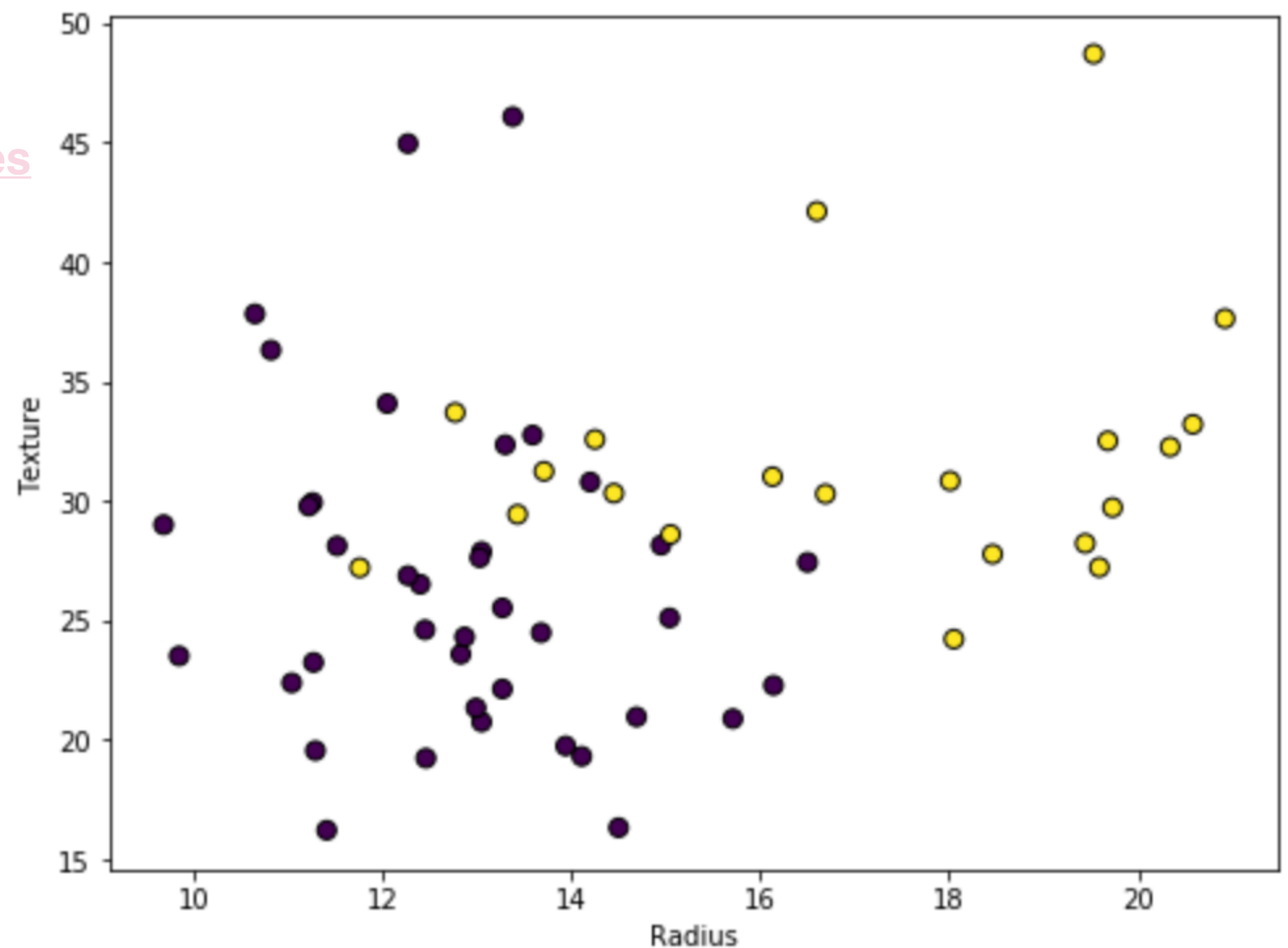
$$d(x_i, x_j) = \sqrt{\sum_{d=1}^D (x_i^d - x_j^d)^2}$$

sum of (squared) differences between the attributes of instances x_i and x_j

The k -nearest neighbors (k -NN) algorithm

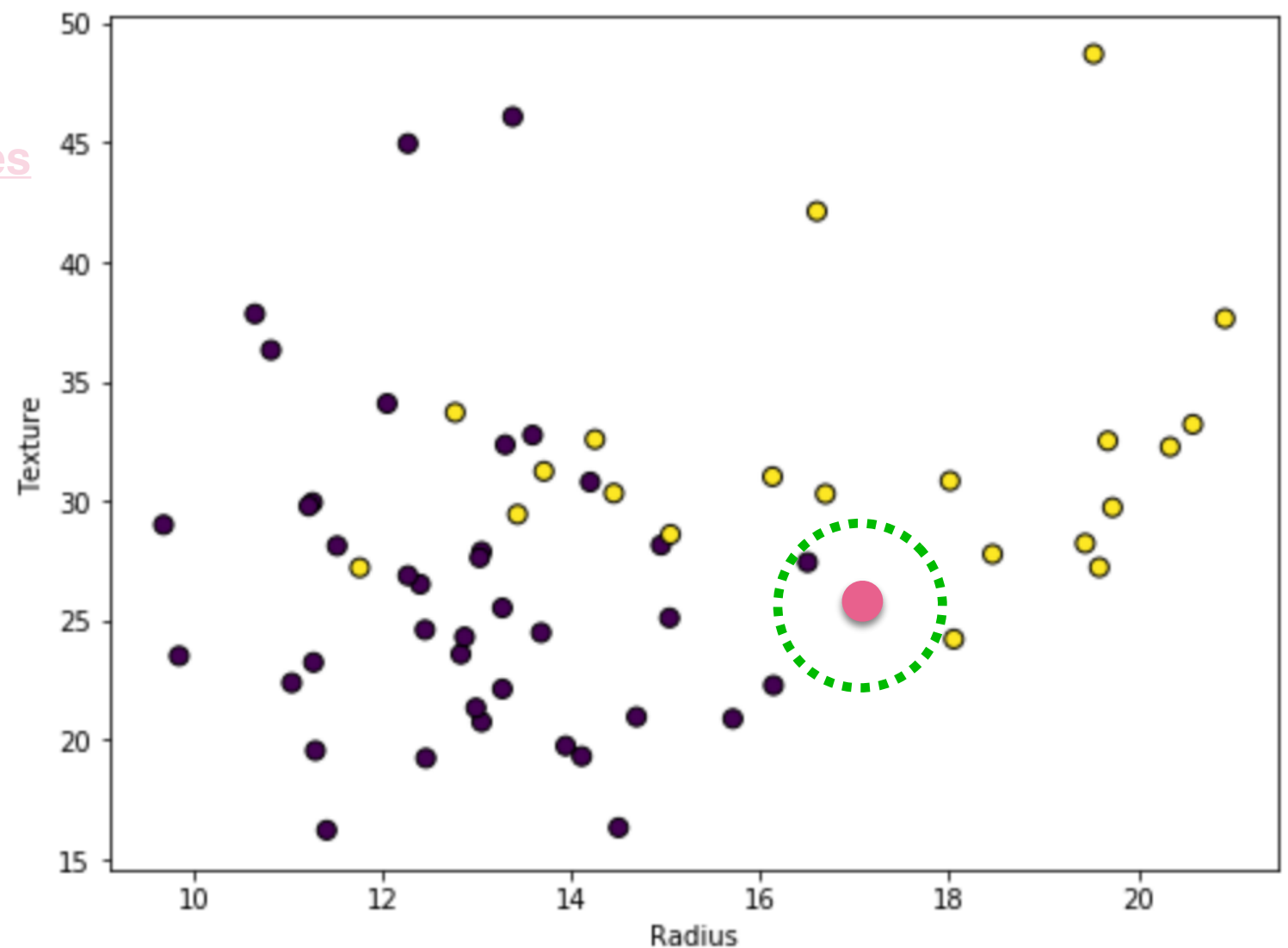
- Instances belonging to a same class are likely to be near in the space of input attributes
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- parameter of the algorithm
- selected by the user
- depends on the application

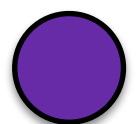


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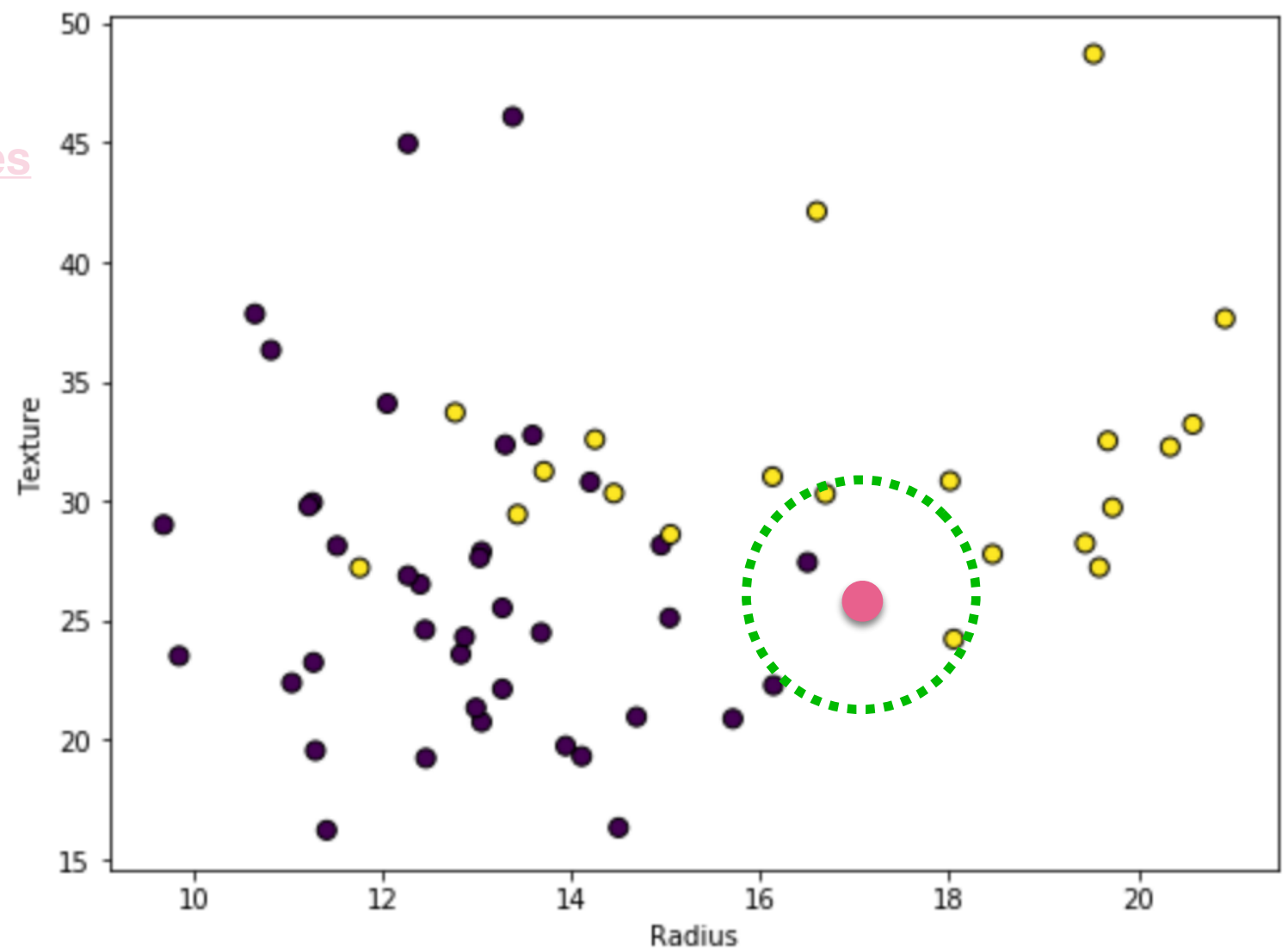


Assume $k=1$
What is the predicted class?

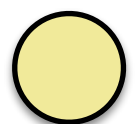


The k -nearest neighbors (k -NN) algorithm

- Instances belonging to a same class are likely to be near in the space of input attributes
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 - parameter of the algorithm
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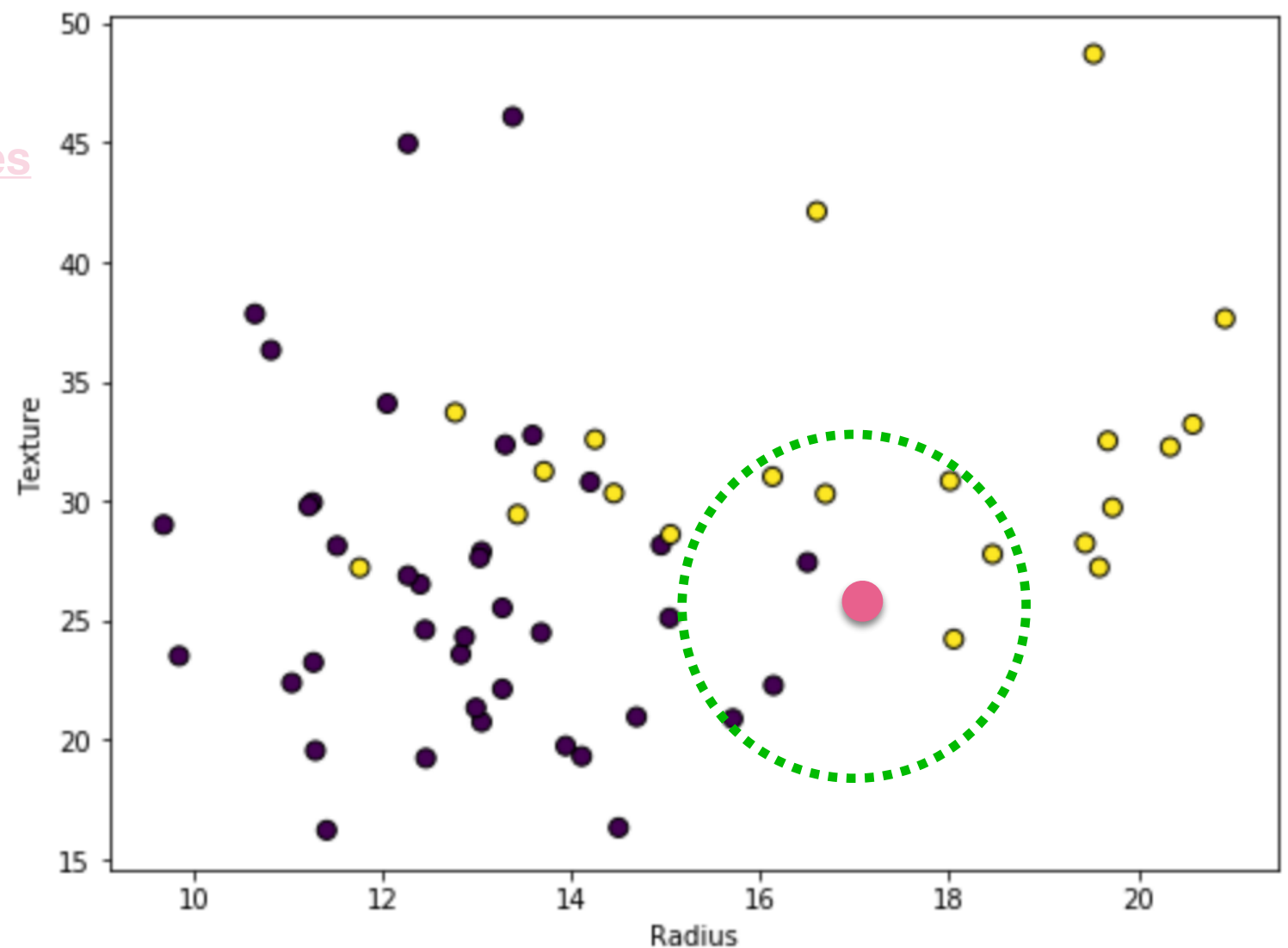


Assume $k=3$
What is the predicted class?

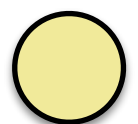


The k -nearest neighbors (k -NN) algorithm

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- Instances belonging to different classes are likely to be far away in the space of input attributes
- To classify a new instance (with unknown class/label) its k nearest instances vote
 - parameter of the algorithm
 - selected by the user
 - depends on the application

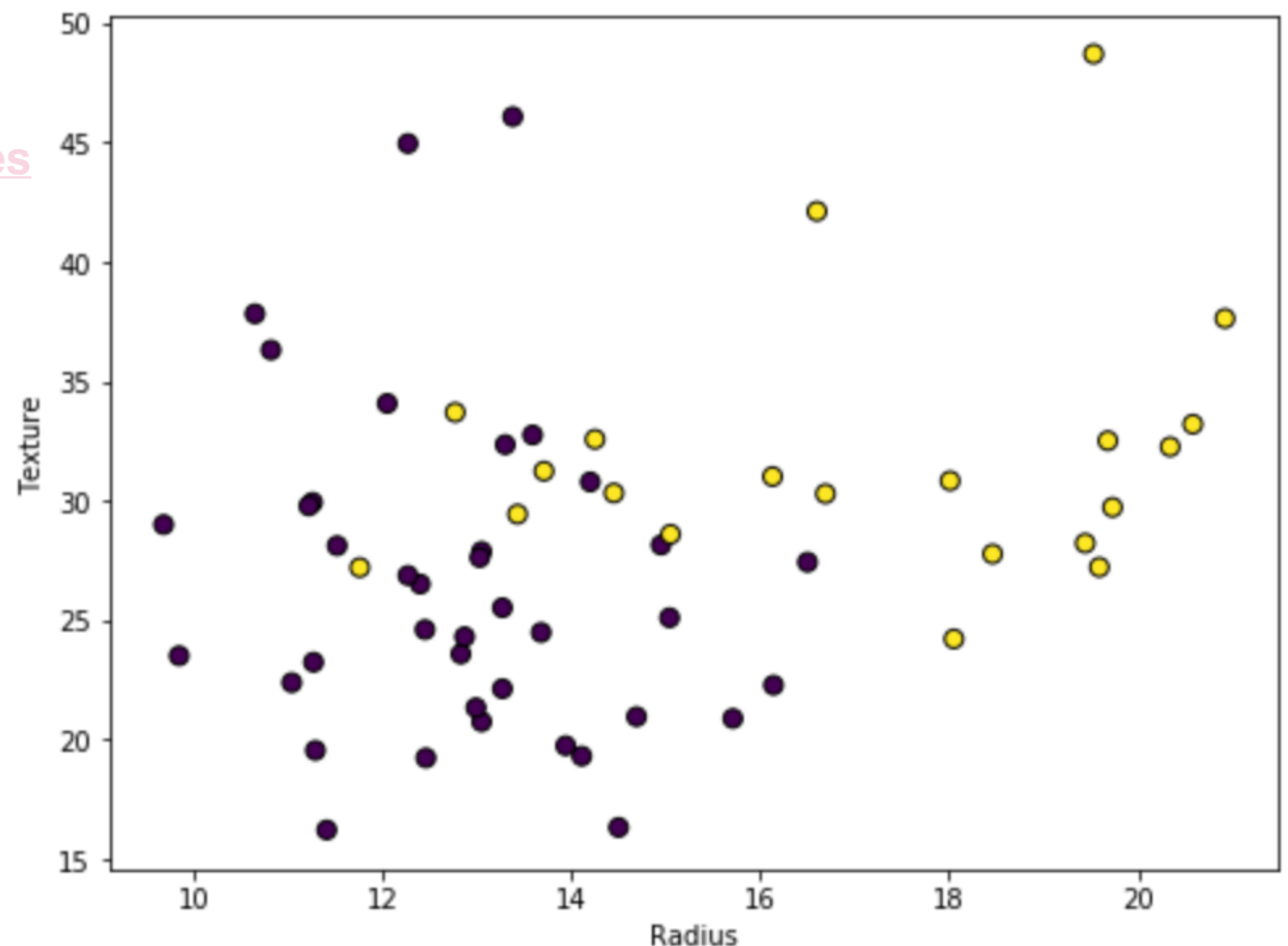


Assume $k=8$
What is the predicted class?



The k -nearest neighbors (k -NN) algorithm

- Instances belonging to a same class are likely to be near in the space of input attributes
- Instances belonging to different classes are likely to be far away in the space of input attributes
- To classify a new instance (with unknown class/label) its k nearest instances vote

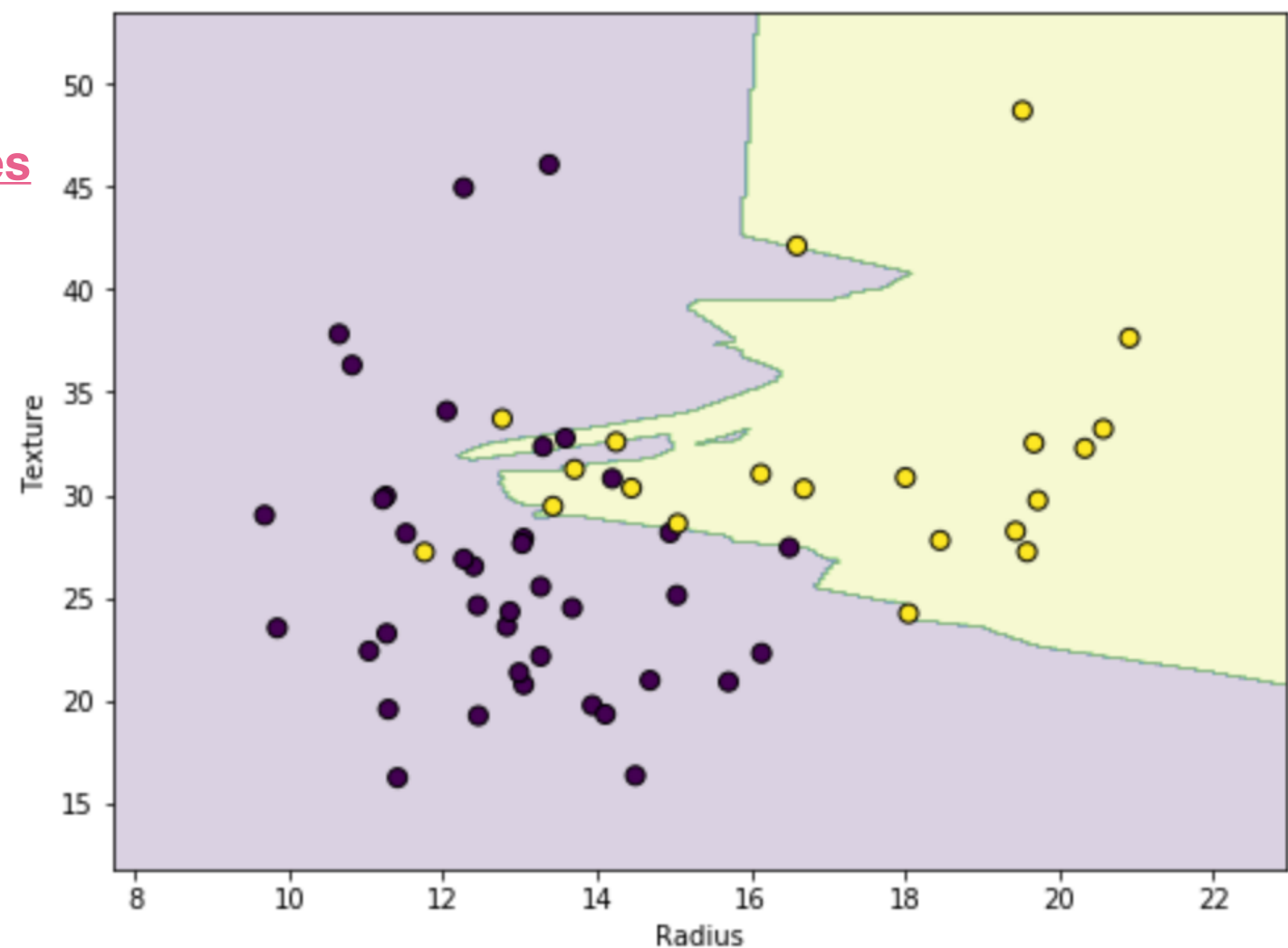


Although extremely simple, the k -NN algorithm
is capable of learning complex non-linear decision boundaries

The k -nearest neighbors (k -NN) algorithm

$k=5$

- Instances belonging to a same class are likely to be near in the space of input attributes
- Instances belonging to different classes are likely to be far away in the space of input attributes
- To classify a new instance (with unknown class/label) its k nearest instances vote

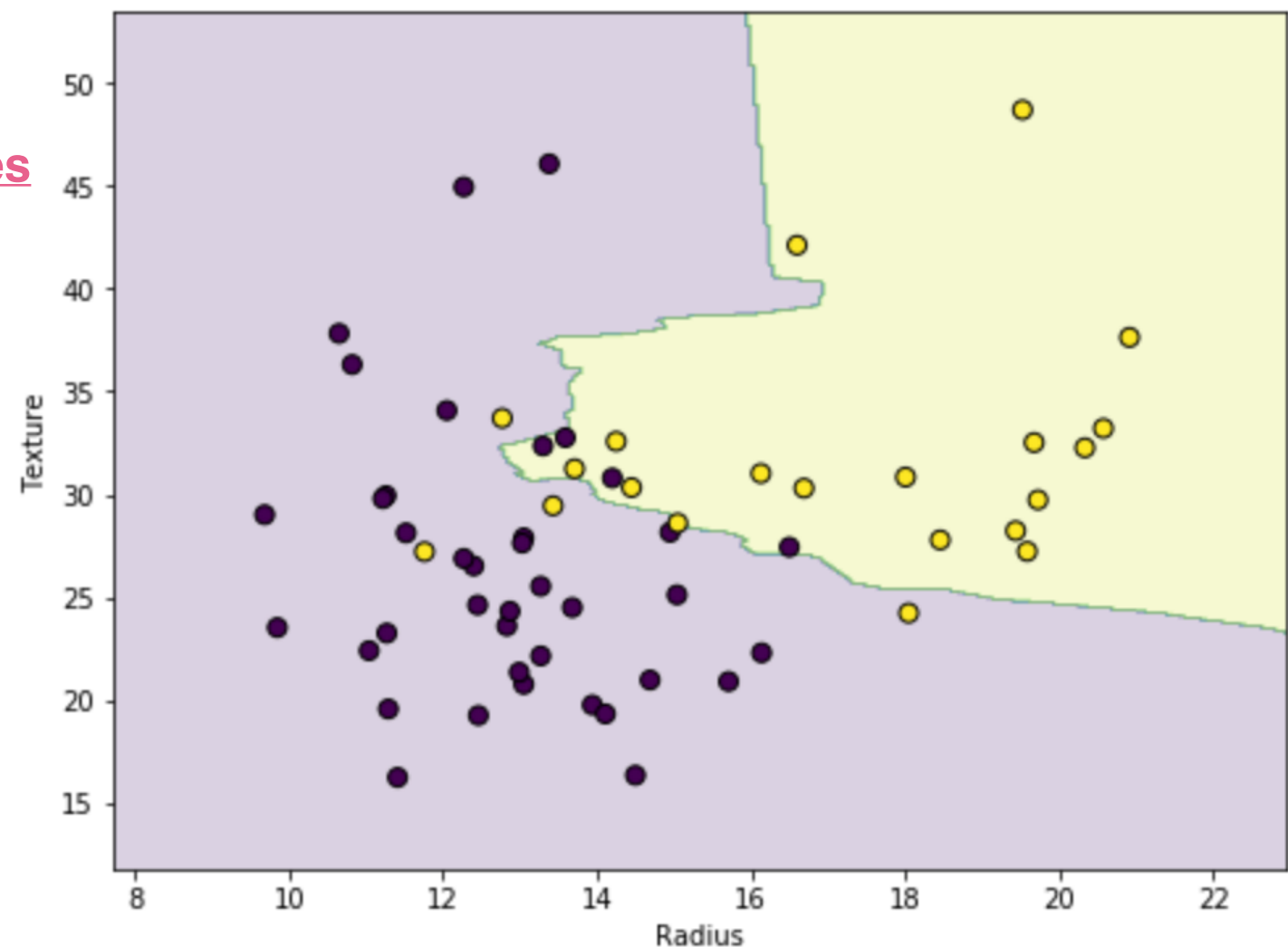


Although extremely simple, the k -NN algorithm is capable of learning complex non-linear decision boundaries

The k -nearest neighbors (k -NN) algorithm

- Instances belonging to a same class are likely to be near in the space of input attributes
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- To classify a new instance (with unknown class/label) its k nearest instances vote

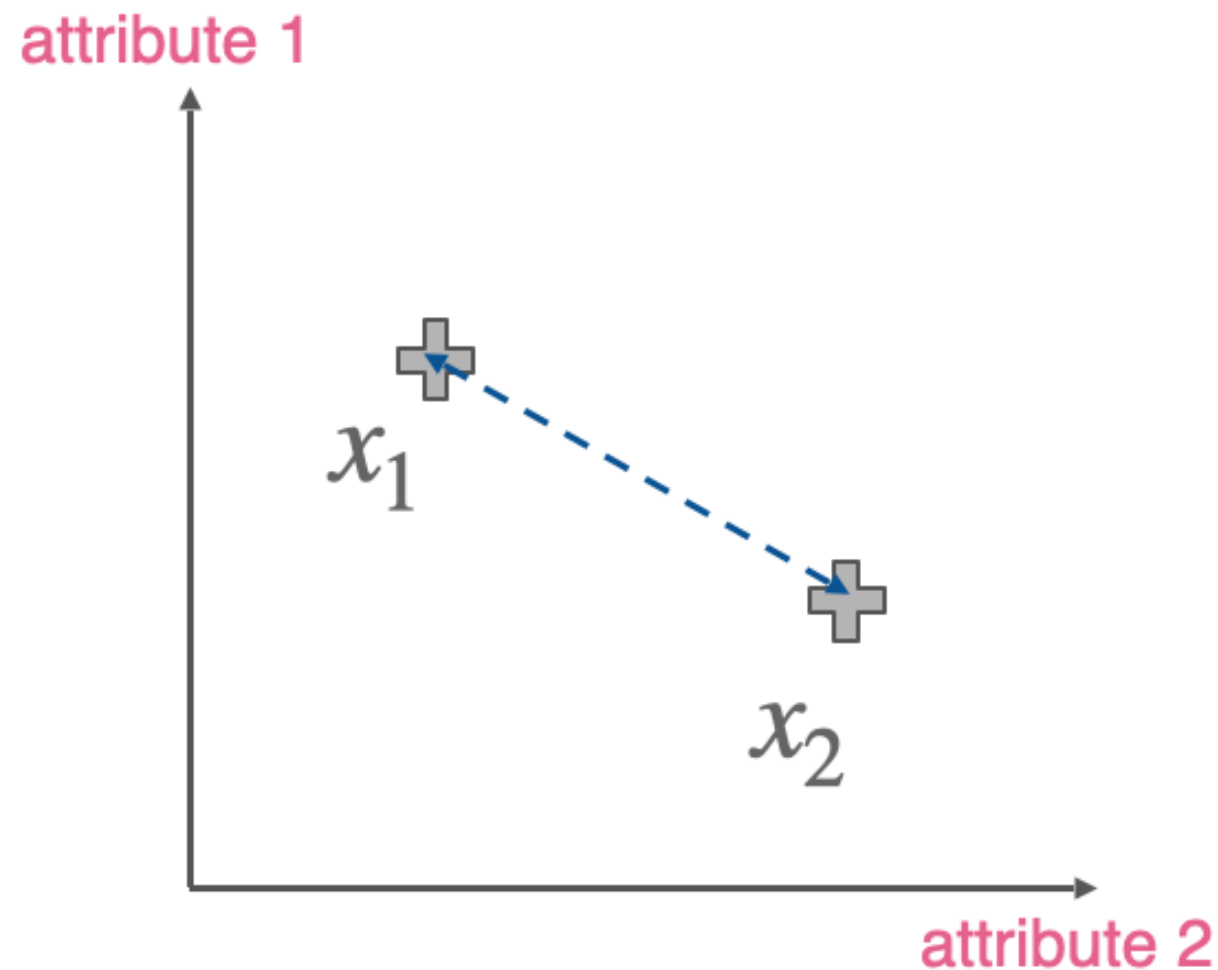
$k=13$



Although extremely simple, the k -NN algorithm
is capable of learning complex non-linear decision boundaries

Nearest neighbors - Euclidean Distance

- Consider, as an example that each instance is described by 2 features/attributes
- Then, each instance in the dataset $\rightarrow$ a point in 2D



Nearest neighbors - Euclidean Distance

- Consider, as an example that each instance is described by 2 features/attributes
- Then, each instance in the dataset → a point in 2D

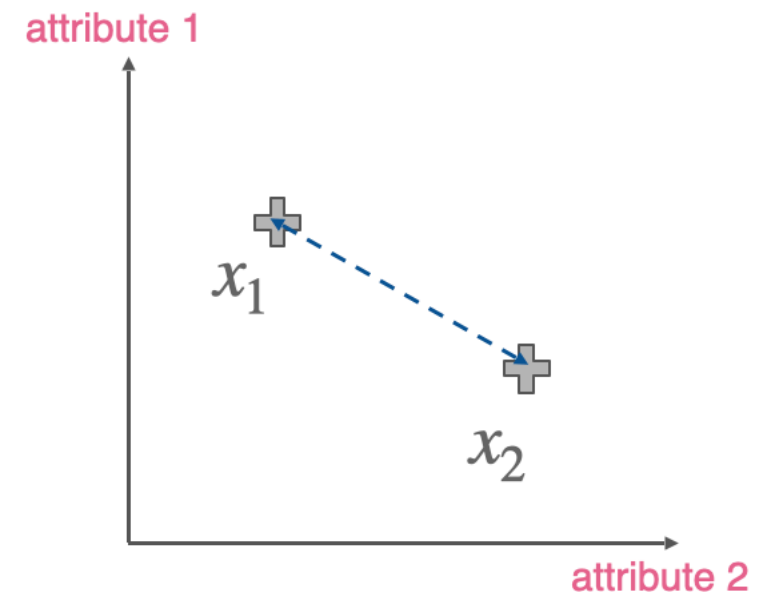
	Attribute 1	Attribute 2
Instance 1 (x_1)	0.5	0.43
Instance 2 (x_2)	1.0	0.57

	Attribute 1	Attribute 2
Instance 1 (x_1)	x_1^1	x_1^2
Instance 2 (x_2)	x_2^1	x_2^2

$$x_1 = [x_1^1, x_1^2] = [\mathbf{0.5}, \mathbf{0.43}]$$

$$x_2 = [x_2^1, x_2^2] = [\mathbf{1.0}, \mathbf{0.57}]$$

Number of attributes
of each instance



$$d(x_i, x_j) = \sqrt{\sum_{d=1}^D (x_i^d - x_j^d)^2}$$

Nearest neighbors - Euclidean Distance

- Consider, as an example that each instance is described by 2 features/attributes
- Then, each instance in the dataset → a point in 2D

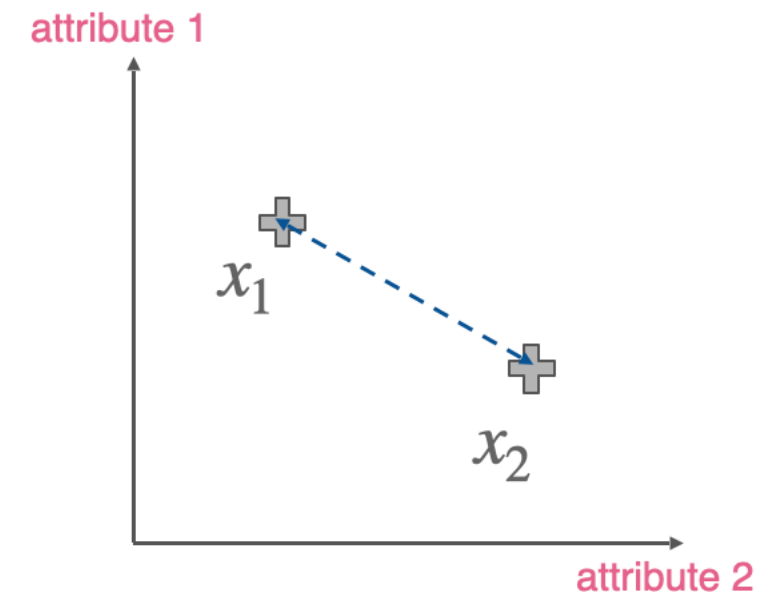
	Attribute 1	Attribute 2
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	Attribute 1	Attribute 2
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$$x_1 = [x_1^1, x_1^2] = [0.5, 0.43]$$

$$x_2 = [x_2^1, x_2^2] = [1.0, 0.57]$$

Number of attributes
of each instance



$$d(x_i, x_j) = \sqrt{\sum_{d=1}^D (x_i^d - x_j^d)^2}$$

$$d(x_1, x_2) = \sqrt{\sum_{d=1}^2 (x_1^d - x_2^d)^2} = \sqrt{(0.5 - 1.0)^2 + (0.43 - 0.57)^2} = 0.5192$$

Nearest neighbors - Manhattan Distance

- Consider, as an example that each instance is described by 2 features/attributes
- Then, each instance in the dataset → a point in 2D

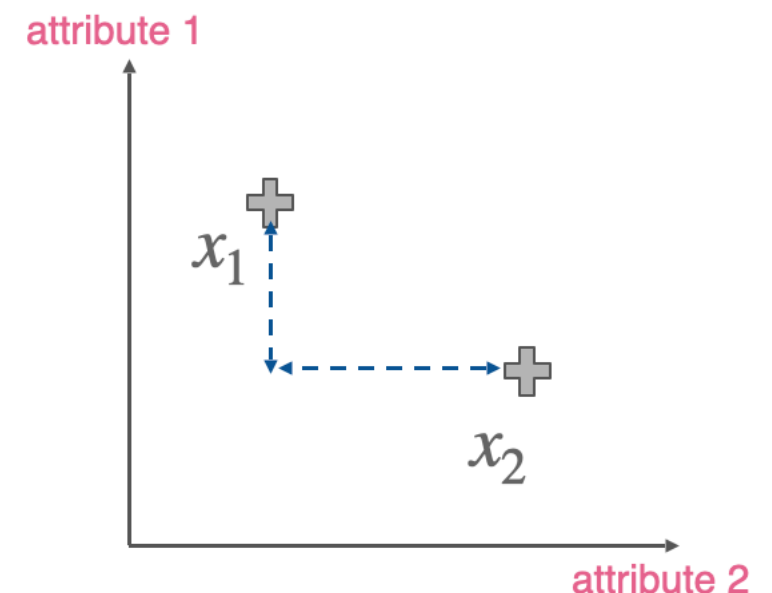
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	Attribute 1	Attribute 2
Instance 1 (x_1)	x_1^1	x_1^2
Instance 2 (x_2)	x_2^1	x_2^2

$$x_1 = [x_1^1, x_1^2] = [\mathbf{0.5}, \mathbf{0.43}]$$

$$x_2 = [x_2^1, x_2^2] = [\mathbf{1.0}, \mathbf{0.57}]$$

Number of attributes
of each instance



$$d(x_i, x_j) = \sum_{d=1}^D |x_i^d - x_j^d|$$

where $|a - b| := \text{abs}(a - b)$

Nearest neighbors - Manhattan Distance

- Consider, as an example that each instance is described by 2 features/attributes
- Then, each instance in the dataset → a point in 2D

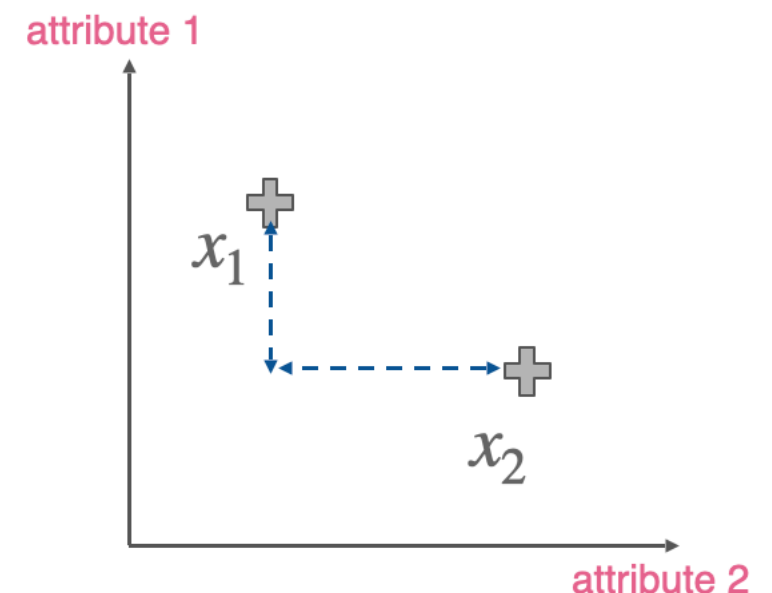
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Number of attributes
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$$d(x_i, x_j) = \sum_{d=1}^D |x_i^d - x_j^d|$$

where $|a - b| := \text{abs}(a - b)$

$$d(x_1, x_2) = \sum_{d=1}^2 |x_1^d - x_2^d| = |0.5 - 1.0| + |0.43 - 0.57| = 0.64$$

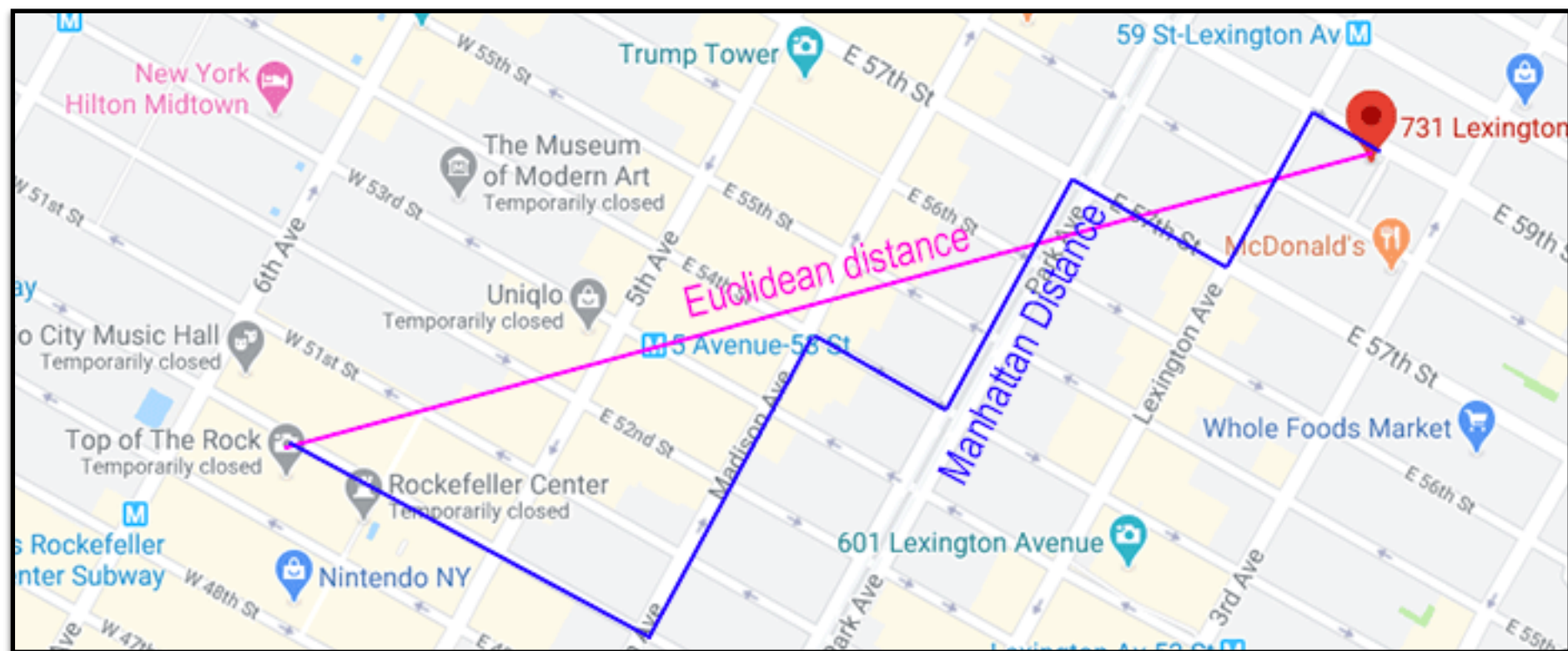
Nearest neighbors - Distance metrics

Euclidean distance

$$d(x_i, x_j) = \sqrt{\sum_{d=1}^D (x_i^d - x_j^d)^2}$$

Manhattan distance

$$d(x_i, x_j) = \sum_{d=1}^D |x_i^d - x_j^d|$$



Nearest neighbors - Distance metrics

- Measuring distances (or “similarity”) between instances might not be easy
 - Are we comparing people according to physical attributes?
 - Are we comparing objects according to their colors?
 - ...

E.g., how to measure the “distance” between two pictures?

picture1 = (pixel1, pixel2, pixel3, pixel4)

picture2 = (pixel1, pixel2, pixel3, pixel4)

d(**picture1**, **picture2**) ?

→ Euclidean?



?

- Many distance metrics exist
 - Which one to use?
 - Depends on the application, type of information stored in each instance, etc.

The k -nearest neighbors (k -NN) algorithm

The k -nearest neighbors (k -NN) algorithm

Input: a dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ with n training instances
an unknown instance, X , that we wish to classify (i.e., what is the label Y of X ?)

Algorithm parameters: a distance metric d
the number of neighbors to use, k

For each training instance, i , in D *(that is, for i from 1 to n)*

 Compute distance $d(X, x_i)$

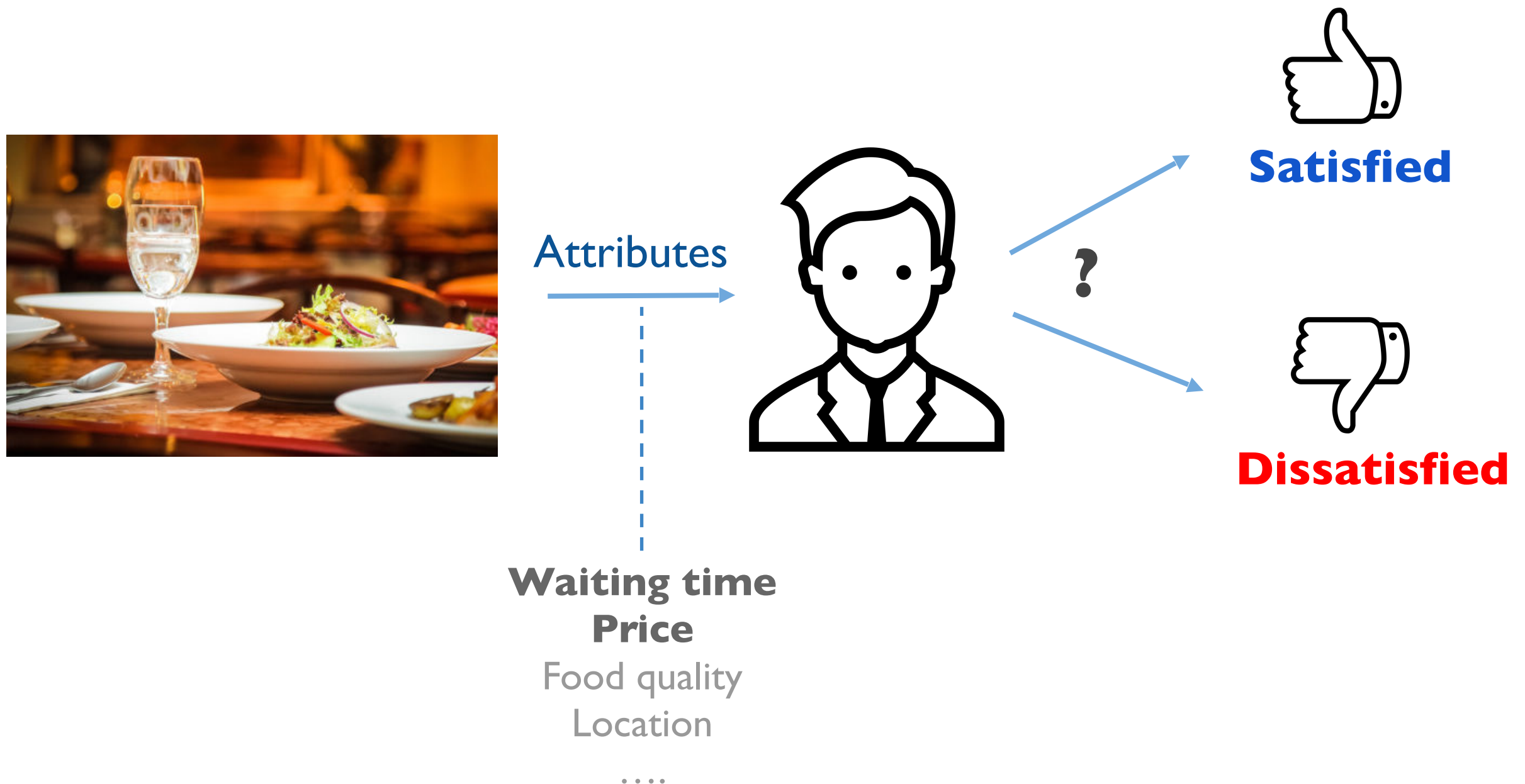
Let C be the set containing the k instances with smallest distances to X

 Let Y be the majority label/class among the instances in C

 Return Y

The 1-nearest neighbors (1-NN) algorithm

- Simplest version of k -NN, with $k = 1$:



The *l*-nearest neighbors (*l*-NN) algorithm

- Example of *l*-NN, assuming attributes have already been normalized

id	WaitingTime	Price	Class
1	0.70	0.40	SAT
2	0.40	0.50	SAT
3	0.61	0.40	SAT
4	0.90	0.70	SAT
5	0.80	0.30	SAT
6	0.60	1.00	SAT
7	0.20	0.60	DIS
8	0.40	0.30	DIS
9	0.61	0.20	DIS
10	0.60	0.43	DIS
11	0.50	0.10	DIS
12	1.00	0.20	DIS

new	0.3	0.7	?
-----	-----	-----	---

The *l*-nearest neighbors (*l*-NN) algorithm

- Example of *l*-NN, assuming attributes have already been normalized

id	WaitingTime	Price	Class	EuclideanDistance
1	0.70	0.40	SAT	0.500
2	0.40	0.50	SAT	0.224
3	0.61	0.40	SAT	0.431
4	0.90	0.70	SAT	0.600
5	0.80	0.30	SAT	0.640
6	0.60	1.00	SAT	0.424
7	0.20	0.60	DIS	0.141
8	0.40	0.30	DIS	0.412
9	0.61	0.20	DIS	0.588
10	0.60	0.43	DIS	0.404
11	0.50	0.10	DIS	0.632
12	1.00	0.20	DIS	0.860

new	0.3	0.7	?
-----	-----	-----	---

The *1*-nearest neighbors (*1*-NN) algorithm

- Example of *1*-NN, assuming attributes have already been normalized

id	WaitingTime	Price	Class	EuclideanDistance
7	0.20	0.60	DIS	0.141
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-----	-----	-----	---

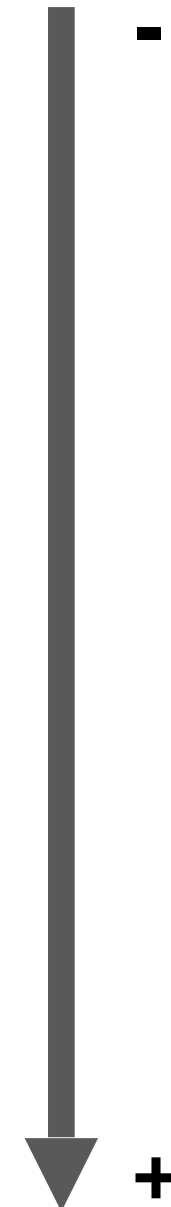


The 1-nearest neighbors (1-NN) algorithm

- Example of 1-NN, assuming attributes have already been normalized

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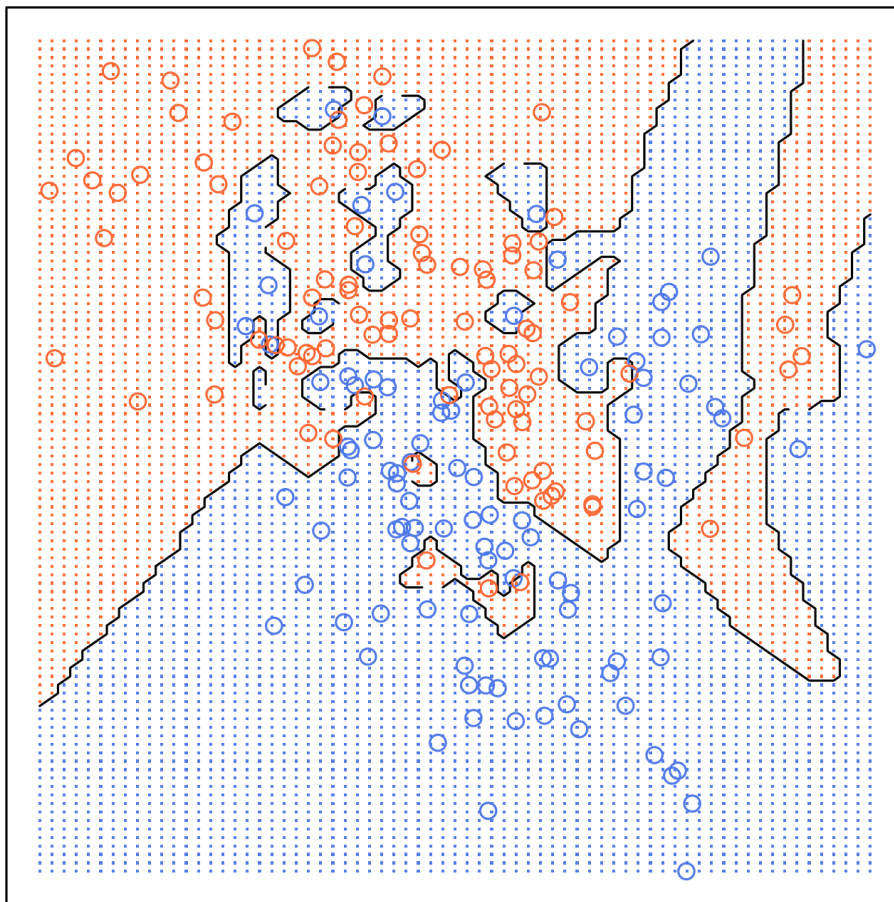
new	0.3	0.7	DIS
-----	-----	-----	-----



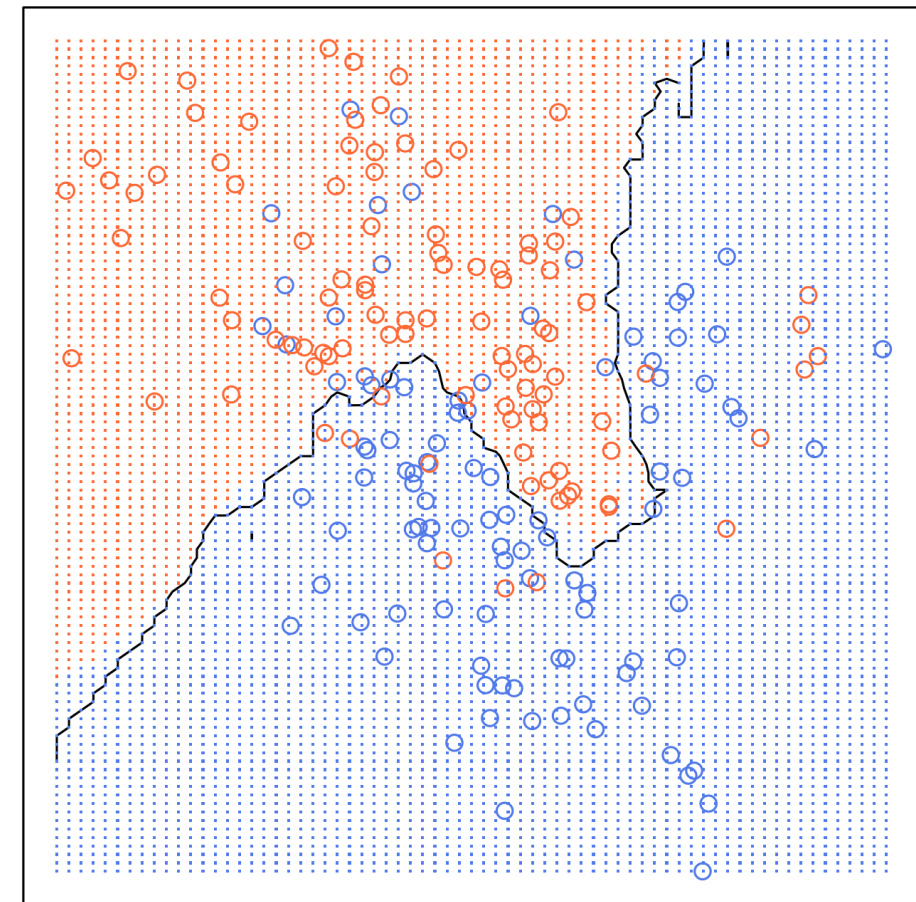
Picking k

- The value of the hyper-parameter k is typically selected empirically
 - it has a direct (and important) impact on the resulting prediction model
- We typically pick an odd number for k to avoid ties

1-nearest neighbours



20-nearest neighbours



The k -nearest neighbors (k -NN) algorithm

The k -nearest neighbors (k -NN) algorithm

Input: a dataset $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ with n training instances
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For each training instance, i , in D *(that is, for i from 1 to n)*

 Compute distance $d(X, x_i)$

Let C be the set containing the k instances with smallest distances to X

 Let Y be the majority label/class among the instances in C

 Return Y

• Pros

- Extremely simple, but yet can learn complex decision boundaries
- No computation cost to “train” the algorithm
 - Just needs access to the dataset — queries it only when actually making predictions

• Cons

- It might be slow to make predictions
 - Compute distance between new instance and possibly millions of instances in the dataset
- Problems with high-dimensional data (many attributes)
 - Intuitively, the distance between any two instances becomes more and more similar
 - Changing the value of one attribute (out of many) has almost no impact on the distance
 - Distance metric “lose their meaning”



Machine Learning

CMPSCI 589

Bruno C. da Silva

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Introduction to Supervised Learning

The *k*-nearest neighbors algorithm

